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Deconstructing the Stablecoin Sandwich

The Cost of Cross-Border Stablecoin Transfers

Prepared by André Reslow, Edona Reshidi, Andrew Usher, Jeanne Verrier, and Marcello Miccoli *

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Deconstructing the Stablecoin Sandwich: The Cost of Cross-Border Stablecoin Transfers

Prepared by André Reslow, Edona Reshidi, Andrew Usher, Jeanne Verrier, and Marcello Miccoli *†

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ABSTRACT: Stablecoins are presented as a low-cost alternative to traditional cross-border payment channels, but evidence to support this claim remains limited. We develop a cost accounting framework that traces the full "stablecoin sandwich'": fiat in, stablecoin transfer, fiat out. We manually collect data on costs from publicly available information across 270 directed corridors covering 18 countries, 29 exchanges, and over 4,000 complete routes. Assuming users select the cheapest options for fiat on- and off- ramps, we estimate that the average cost of a $200 transfer is around 2 percent across all corridors, below the G20 target of 3 percent and cheaper than traditional remittances on average. However, in roughly one-third of corridors where granular information on costs through traditional routes is available, the cheapest traditional provider offers a lower cost than the cheapest stablecoin route. Trading fees and fiat rail (on- and off-ramp) fees account for the majority of total costs, while the stablecoin premium—the deviation of local exchange prices of stablecoins from benchmark exchange rates—is the primary driver of cross-corridor variation, pushing costs below zero in some corridors and above 8 percent in others.

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1 Introduction

Cross-border payments are slow, expensive, opaque, and unevenly accessible. The global average cost of sending $200 was 6.36 percent in the third quarter of 2025, more than double the G20 target of 3 percent (World Bank, 2025). The latest FSB progress report on cross-border payments notes that costs remain above target in many cases, underscoring that progress has not yet translated into tangible improvements for end-users at the global level (Financial Stability Board, 2025). In many corridors, especially those serving low-income or geographically remote communities, costs are substantially higher.

Stablecoins have emerged as a potential alternative to traditional cross-border payment rails. Stablecoin cross-border flows were estimated at approximately $6.3 trillion in 2025—still only a small fraction of the roughly $1 quadrillion global market for traditional cross-border payments—but have been growing rapidly in recent years (International Monetary Fund, 2025, 2026). Dollar-denominated stablecoins, principally USDT and USDC, can in principle be purchased on an exchange in one country, transferred across a blockchain network, and sold for local currency on an exchange in another country, bypassing the correspondent banking chain that underpins most traditional remittances. A growing body of industry commentary and policy discussion has pointed to stablecoins as a potentially cheaper and faster channel for cross-border transfers. Yet, data is lacking.

This paper fills the gap by providing the first systematic, end-to-end measurement and decomposition of the cost of cross-border stablecoin transfers. We develop a cost accounting framework that traces a stablecoin transfer through five steps: fiat deposit (on-ramp), stablecoin purchase, on-chain transfer, stablecoin sale, and fiat withdrawal (off-ramp). We then estimate costs across 270 directed corridors covering 18 countries, 29 exchanges, two stablecoins (USDT and USDC), and five blockchain networks, yielding 4,005 complete transfer routes. We manually collect this information by relying on publicly available information provided by crypto exchanges. To our knowledge, this is the first paper to measure the full cost stack of stablecoin-based cross-border transfers at this scale.

The claim that stablecoins reduce transfer costs often rests on the low cost of the blockchain transfer itself. Our results show that this focus is incomplete. A remittance sender does not simply transfer tokens on a blockchain. The sender must first convert domestic currency into a stablecoin through a crypto exchange; the recipient must then convert the stablecoin back into local currency and withdraw the proceeds into the banking system. The total cost reflects frictions across this entire transaction chain, not just the blockchain leg.

Specifically, we focus on a retail remittance-type transaction in which fiat currency is exchanged for a U.S. dollar stablecoin at the sending side, transferred on-chain between exchanges, and then redeemed into local currency and withdrawn to a bank account at the

receiving side. This "stablecoin sandwich" structure, in which fiat enters and exits through traditional payment rails while only the cross-border leg is executed on a blockchain, is the dominant architecture for exchange-based stablecoin cross-border payments.

To facilitate interpretation of the results, we distinguish the main cost components of a stablecoin transfer, as illustrated in Figure 1. Fiat deposit fees are charged when funds are deposited into the origin exchange, while fiat withdrawal fees are charged when funds are withdrawn from the destination exchange. When users purchase stablecoins, exchanges charge a proportional trading fee. The remaining balance is then exchanged for stablecoins at the price quoted to buyers (the ask price), which determines how many stablecoins the user receives. Rather than treating the purchase price as a single number, we decompose it into the premium and the spread. The premium captures deviations of the stablecoin price from the benchmark exchange rate, while the spread captures the difference between the prices at which users buy and sell stablecoins. Finally, crypto withdrawal fees are incurred when stablecoins are withdrawn from the origin exchange for transfer to the destination exchange, while crypto deposit fees are incurred when the transferred stablecoins are deposited into the destination exchange.

Our main findings are as follows. The average cost of a nominal $200 stablecoin transfer across 270 corridors is 2.18 percent, with a median cost of 1.99 percent and substantial variation from −2.18 to 8.03 percent. Trading fees are the largest cost component (0.85 percent on average), followed by fiat rail fees (0.67 percent) and crypto transfer costs (0.48 percent). The premium and spread component averages only 0.18 percent but is the most variable, ranging from −3.07 to 4.51 percent, and the premium is the primary driver of cross-corridor cost heterogeneity. The 15 cheapest corridors have negative average costs, driven by large destination-side premiums, while the 15 most expensive corridors average 6.88 percent, driven by fiat rail fees and premiums. Blockchain transfer fees, often highlighted as the main source of cost savings, account for less than one quarter of average total costs. This suggests that the main sources of stablecoin transfer costs lie in domestic exchange infrastructure, stablecoin pricing frictions, and fiat payment rail access.

The cost estimates at the corridor level weight routes by observed number of transactions across exchanges and by global on-chain market shares across coins and blockchain networks, based on data from private providers. The average cost is then computed as the simple average across all corridors. For fiat deposit and withdrawal methods, no equivalent usage data exist to determine which options users actually select; we therefore use the cheapest available method at each exchange as the baseline, which in practice most commonly corresponds to standard bank transfers.

The role of the premium can be illustrated using Korea as an example. Stablecoins in Korea tend to trade at a persistent premium relative to the benchmark exchange rate—the so-called "Kimchi premium" (Choi et al., 2022; Kang et al., 2025)—meaning that buyers of stablecoins in Korea pay more Korean won per stablecoin than the benchmark exchange

rate.[1] For a sender in Korea, this premium raises the cost of acquiring stablecoins at the origin, increasing total transfer costs. For a receiver in Korea, however, the same premium works in the opposite direction: the recipient sells stablecoins into a market where they command a higher local price, receiving more Korean won per stablecoin than the benchmark would imply. This can reduce total transfer costs substantially, and in corridors where the destination-side premium is large enough to offset all fees, the measured transfer cost becomes negative. Importantly, a negative cost does not imply that users are paid to remit or that the transfer technology itself generates value. Rather, it reflects a pricing distortion relative to the benchmark exchange rate, potentially arising from factors such as market segmentation, strong local demand for dollar-denominated assets, or limits to arbitrage.

When comparing to traditional remittance channels, we find that stablecoin transfers are cheaper on average. In the 33 corridors where both stablecoin costs and Remittance Prices Worldwide data are available, the average stablecoin cost is almost 5 percentage points below the average cost of traditional providers. However, the comparison is more nuanced when focusing on the cheapest available route within each corridor: when comparing the cheapest stablecoin route with the cheapest traditional provider in each corridor, traditional options remain less expensive in approximately one-third of corridors. This suggests that stablecoins offer a cost advantage in the typical case but do not uniformly dominate traditional channels, particularly where low-cost traditional providers already operate. The two comparisons answer different questions. The average corridor comparison aggregates routes using observed market activity weights and therefore reflects costs across the routes used in the market, rather than assuming that all users search and select the minimum-cost route. By contrast, comparing the cheapest stablecoin route with the cheapest traditional provider within each corridor identifies the lowest-cost option available to users.

We examine the sensitivity of our findings to alternative assumptions and scenarios. Mean corridor costs are stable across alternative weighting schemes for aggregating routes into corridors, suggesting that the baseline results are not primarily driven by the particular weights chosen. When median- or highest-cost methods are used instead of the cheapest available fiat payment method, mean corridor costs rise to 3.04 and 4.07 percent respectively, confirming that the choice of fiat rail has a first-order effect on total costs. While we cannot systematically evaluate this for all corridors, with increased cost methods, speed also increases. Transfer size also has a meaningful effect: the mean corridor cost falls from 4.12 percent at \$50 to 1.54 percent at \$5,000, reflecting the fixed nominal component of crypto withdrawal and fiat fees. This implies that stablecoin transfers are

[1] We note that after the sample considered in our paper, specifically the first half of 2026, the premium narrowed and even reversed. A complete discussion of the drivers and causes of the premium on stablecoins over benchmark rates is beyond the scope of this paper.

relatively more expensive for small remittances.

These results should be interpreted in light of the scope of our cost measure. The paper focuses on measurable transaction costs faced by users. Other factors relevant to users, including transfer speed and other frictions associated with search, onboarding, and authentication costs, cannot be systematically observed or inferred from our data and are therefore not included in our estimates, although they may affect the relative cost and attractiveness of different transfer methods. For instance, our assumption of the cheapest available method for fiat deposits typically favors a standard bank transfer, which is generally slower than a credit card transfer but significantly cheaper. Speed considerations may also influence blockchain choices, as higher fees can incentivize faster transaction processing. Similarly, identifying the lowest-cost stablecoin route may entail search costs, while some fees may only become observable after onboarding or user authentication. Beyond these user-side costs and frictions, our analysis does not account for other considerations that are relevant to users but do not constitute transaction costs, such as counterparty risk. Our analysis also focuses exclusively on costs faced by users and does not consider costs borne by intermediaries. Regulatory and compliance costs incurred by exchanges and other intermediaries are therefore outside the scope of the paper.

Our analysis is also limited to the centralized-exchange-to-centralized-exchange transfer structure. We do not study peer-to-peer transfers, including those facilitated by centralized exchanges, decentralized exchange (DEX) conversions, stablecoin-to-stablecoin swaps using DeFi protocols, or use cases in which the recipient holds the stablecoin rather than converting it to fiat. While these alternatives exist, the centralized-exchange-to-centralized-exchange path represents the most directly comparable structure to traditional remittance channels and is the one for which systematic fee and pricing data can be collected. This choice does not imply any view on which model is most likely to prevail in the future. We also do not examine the underlying sources of the observed cost differences between stablecoin-based and traditional transfer methods. These differences may reflect, among other factors, differences in regulatory requirements, technology, or the processes and infrastructure involved in executing transfers. Finally, our estimates provide a snapshot of transfer costs under the fees and market conditions prevailing at the time of data collection, rather than their evolution over time.

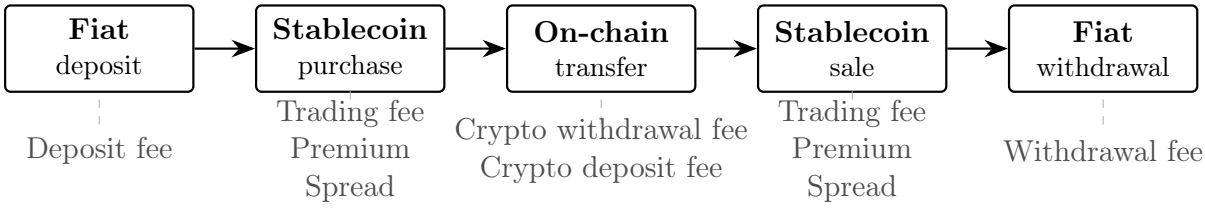


Figure 1: Cost Components at each Stage of a Stablecoin Transfer

Related literature: This paper contributes to a rapidly growing literature on stablecoins and cross-border payments. The closest paper to ours is Du et al. (2026), who compare the cost of cross-border payments across three methods: correspondent banking, fintech money transmitters, and stablecoins. To facilitate comparisons across these different rails, they construct a harmonized all-in cost measure consisting of FX markups and explicit fees, using intraday stablecoin-fiat transaction data from CoinDesk and Bloomberg benchmark rates. They find that stablecoin rails reduce certain settlement frictions but do not eliminate intermediation, and that stablecoins exhibit cost advantages primarily in capital-controlled or segmented FX markets. Our analysis provides a more granular and systematic measurement of the end-to-end cost of stablecoin transfers. We trace a transfer through five stages and measure ten distinct fee and pricing components, allowing us to identify both the total cost of using the stablecoin rail and the sources of that cost. This distinction is important because similar all-in costs can arise from economically distinct sources, including domestic fiat rails, exchange intermediation, blockchain transfers, and local stablecoin pricing. Our decomposition therefore reveals which components account for the average level of costs and which drive variation across corridors, and helps identify which margins are most relevant for efforts to lower end-user costs. We find that fiat rail and trading fees account for the majority of average transfer costs, whereas stablecoin premiums are the main source of cross-corridor variation. Such breakdown of costs allows more analysis by policymakers to determine the sources of costs for transfers.

A recent paper by Di Iorio et al. (2026) uses a mystery-shopping exercise to execute USDC transfers across ten corridors linking Italy with five other countries. They find that total costs range from 0.30 percent to nearly 9 percent for a $200 transfer, with substantial variation in execution times. They also find that the on-chain transfer accounts for only a small share of total costs, with domestic payment infrastructure playing an important role in overall transaction efficiency. While their approach provides direct evidence from executed transactions, our analysis provides substantially broader and more systematic coverage of stablecoin transfer costs. Our dataset spans 270 directed corridors across 18 countries, 29 exchanges, two stablecoins, and five blockchain networks, yielding 4,005 complete transfer routes. This coverage allows us to examine how both the level and composition of costs vary systematically across countries, transfer directions, exchanges, stablecoins, and blockchain networks, and to identify the sources of cross-corridor cost heterogeneity.

The premium links our paper to a second strand of literature on stablecoin pricing and parity deviations. Aldasoro et al. (2026) document persistent gaps between the cost of acquiring dollars through stablecoins and through traditional FX spot markets across 27 currencies. They estimate that a 1 percent exogenous increase in net stablecoin inflows raises parity deviations by 40 basis points, depreciates the local currency, and widens CIP deviations. These parity deviations correspond closely to the premium component in our

cost framework, and our results show that this component is the largest source of cost variation across corridors. The connection between crypto premia and capital controls has broader roots in the literature: Graf von Luckner et al. (2023) and Graf von Luckner et al. (2024) document that Bitcoin has become an important channel for remittances and capital flight in emerging markets precisely where official exchange rates are most constrained, consistent with the premium patterns we observe in our sample.

A third strand measures cross-border stablecoin and crypto flows. Cardozo et al. (2024), Auer et al. (2025), and Reuter (2025, 2026) develop methods to estimate the geographic distribution and drivers of these flows, documenting large cross-border volumes and links to transactional motives, remittance costs, and global push factors. Our paper complements this literature by measuring the costs of using stablecoins for cross-border transfers.

The remainder of the paper is organized as follows. Section 2 describes the market structure of stablecoin-based transfers. Section 3 presents the cost accounting framework. Section 4 describes the data collection and estimation methodology. Section 5 reports the results. Section 6 presents sensitivity checks. Section 7 concludes.

2 Market Structure

This section describes the institutional structure through which stablecoin-based transfers take place.[2] The central feature of this structure is vertical separation: issuers do not generally connect directly to end users. Instead, stablecoins are issued in a primary market, where only a small set of approved institutional counterparties can mint and redeem tokens directly with the issuer, and then circulate in a secondary market, where users buy, sell, and transfer stablecoins through exchanges and other intermediaries. This market structure, illustrated in Figure 2, means that the costs and risks of stablecoin transfers depend not only on the issuer, but also on the intermediaries and infrastructures that connect issuers to users. Each stage of the transaction chain generates a distinct cost component—fiat deposit and withdrawal fees, exchange trading fees, blockchain transfer costs, and market pricing frictions—whose relative importance is quantified in Section 5.

Issuers and the primary market. Stablecoin issuers create and redeem stablecoins and manage the reserve assets that support their value relative to a reference asset. In the current market, issuance is concentrated around a small number of U.S. dollar-denominated stablecoins, especially USDT, issued by Tether, and USDC, issued by Circle.[3] This paper focuses on USDT and USDC because together they account for roughly

[2]For an overview of traditional cross-border payment market structures see CPMI (2018).

[3]USDT and USDC are the two largest U.S. dollar-pegged stablecoins and account for a significant share of stablecoin activity in emerging market and developing economies (International Monetary Fund,

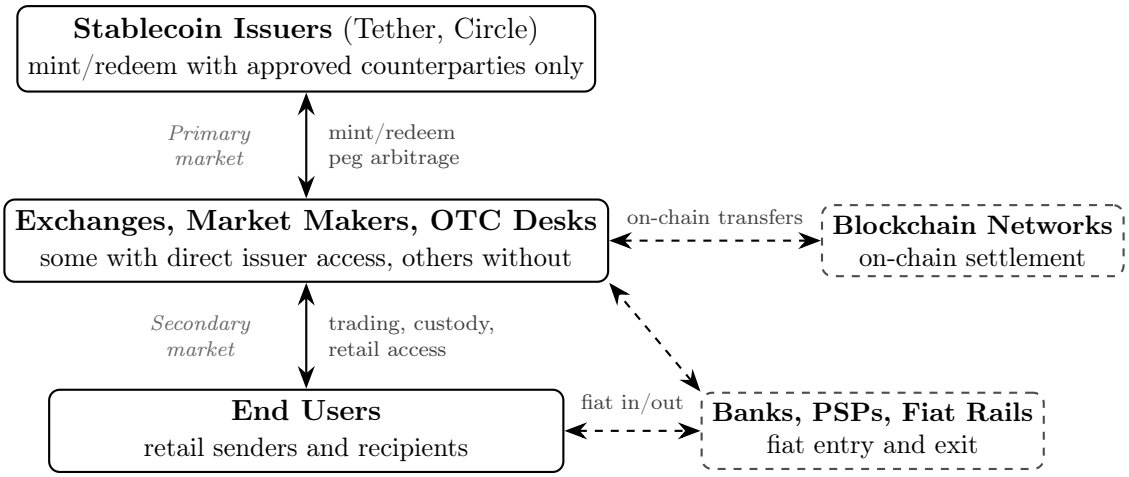


Figure 2: Simplified market structure of stablecoin-based transfers.

95 percent of on-chain stablecoin transaction volume.

Access to direct issuance and redemption is also concentrated. Issuers restrict primary-market access to a narrow set of approved counterparties—typically large exchanges, trading firms, market makers, and OTC desks—that have been onboarded through the issuer's compliance process.[4] Most users cannot redeem stablecoins directly with the issuer; instead, they trade on the secondary market (Watsky et al., 2024). Evidence on Tether suggests that in an average month only a small number of agents redeem USDT for cash through the primary market (Ma et al., 2025).

The issuer business model shapes the broader fee structure. Both Tether and Circle earn the bulk of their revenue from interest on reserve assets—primarily U.S. Treasury securities—rather than from transaction fees. In some cases, issuers share this reserve income with key distribution partners. Under a 2023 collaboration agreement, Circle shares reserve income with Coinbase, its largest distribution partner and an equity stakeholder: Coinbase receives all reserve income on USDC held on its platform and half of income earned elsewhere.[5] On the Tether side, Tether Limited is owned by iFinex, which also operates the Bitfinex exchange, representing a different form of vertical linkage between issuer and trading platform. Such arrangements can lead to differentiated user-facing fees—for example, an exchange receiving reserve-income sharing may charge lower trading fees on USDC than on USDT.

2026)

[4]Both Tether and Circle operate closed issuance and redemption arrangements: stablecoins are minted and redeemed through the issuer, but direct access is limited to eligible and verified customers. See, for example, Circle's USDC Terms of Use.

[5]In 2024, distribution costs paid to Coinbase totaled $908 million, or roughly 54 percent of Circle's total revenue. See Circle Internet Group, Inc., Form S-1/A, filed with the U.S. Securities and Exchange Commission, May 27, 2025.

The secondary market. The secondary market is where most users access stablecoins. It consists of user-facing intermediaries and liquidity providers. User-facing intermediaries—such as exchanges, brokers, wallet providers, custodians, and payment apps—allow users to buy, sell, hold, and transfer stablecoins. These include large global exchanges such as Binance, Coinbase, Kraken, and Crypto.com, as well as local or regional platforms. Liquidity providers—such as market makers and OTC desks—support trading by supplying inventory, quoting prices, and connecting liquidity across venues.

For the cross-border transfer use case studied in this paper, exchanges are especially important because they connect stablecoins to local fiat currencies. A user who wants to acquire USDT or USDC will typically need an exchange, broker, or wallet provider that supports both the relevant stablecoin and a fiat payment method in the local currency. The intermediary may provide the stablecoin from its own inventory, source liquidity from market makers or OTC desks, or trade through other venues. The fees, spreads, liquidity, and operational reliability of these access points therefore shape the effective cost and feasibility of stablecoin-based transfers.

Concentration in this layer differs from issuer concentration. It depends not only on the number of exchanges, but also on where liquidity, fiat access, trading pairs, user balances, and supported stablecoin-network combinations are concentrated. For cross-border transfers, this matters because a route is feasible only when the origin and destination access points support the same stablecoin and blockchain network, as well as the relevant fiat on- and off-ramps. As a result, the set of feasible routes may be much narrower than the number of exchanges alone would suggest.

Supporting infrastructure. Two additional infrastructures support the stablecoin transaction chain and generate further cost components. First, *blockchain networks* provide the settlement and recordkeeping infrastructure on which stablecoins are issued and transferred. In the cross-border use case studied in this paper, stablecoin transfers can take place either within an intermediary's internal ledger or on-chain. If two users hold accounts at the same exchange or hosted wallet provider, the platform can debit the sender's balance and credit the recipient's balance internally; no blockchain transaction takes place, and the transfer is not visible in on-chain data. By contrast, an on-chain transfer occurs when stablecoins move between blockchain addresses and the transaction is recorded on the relevant blockchain network.

Not all stablecoins are supported on all chains. For example, USDT is widely used on Tron and Ethereum, while USDC is widely used on Ethereum and is also available on networks such as Solana and Base. Network support is therefore part of the market structure: the same stablecoin can have different costs and liquidity depending on the blockchain used. The blockchain also affects the feasibility of a route, since the sender's and recipient's exchanges must support the same stablecoin-network combination. In

practice, users often access blockchain networks through exchanges, which choose which networks to support and set crypto withdrawal and deposit fees. These fees may reflect underlying network fees, but can also include exchange-level markups.

Second, *fiat on- and off-ramps*—banks, payment service providers, and local payment methods—enable users and intermediaries to enter and exit the stablecoin ecosystem. Users need a bank account, payment card, or local payment method to convert fiat currency into stablecoins, and a similar arrangement to convert back. Stablecoins therefore do not eliminate the role of traditional payment infrastructure; rather, banks and payment providers remain essential at the entry and exit points of the transaction chain. Concentration at this layer can arise when only a limited number of banks, payment providers, or local exchanges are willing or able to provide fiat access to stablecoin users and intermediaries. This can make stablecoin transfers sensitive to banking relationships, local payment infrastructure, and the availability of fiat liquidity in the relevant currency—factors that vary substantially across the countries in our sample.

A simple transaction example illustrates the interaction of these layers (Figure 3). A sender first enters the ecosystem by depositing local currency at an exchange. The exchange then converts the funds into a dollar stablecoin, either from its own inventory or through external liquidity providers. The sender transfers the stablecoin to the recipient. If the recipient uses an external wallet, the transfer is recorded on-chain. If both parties use the same platform, the transaction may be recorded only on the platform's internal ledger. The recipient can then hold the stablecoin, transfer it onward, or sell it for local currency through an exchange, broker, or payment provider. This type of cross-border structure has been described as a "stablecoin sandwich" in which fiat money is converted into a dollar stablecoin, transferred, and then converted back into fiat money.

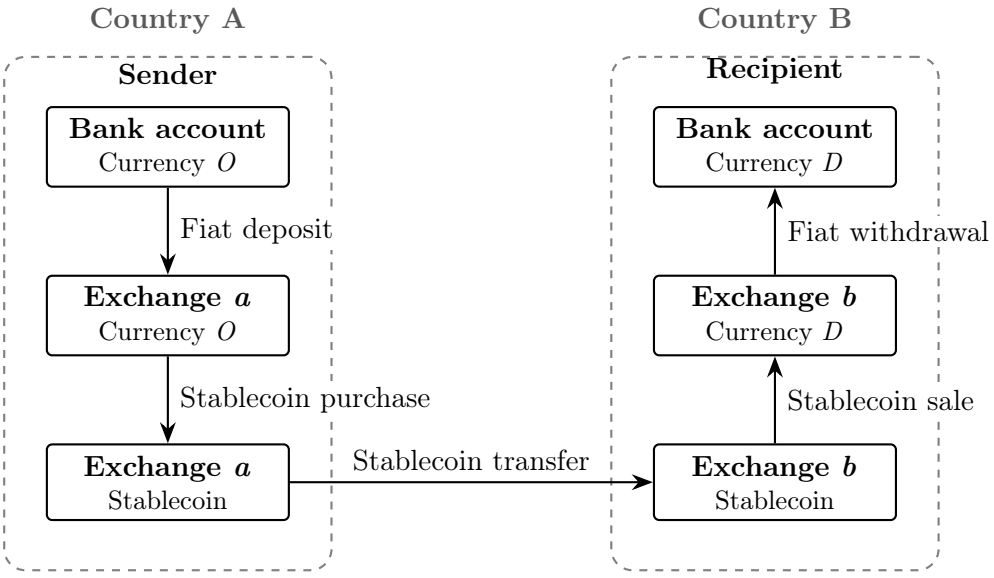


Figure 3: Stylized stablecoin sandwich transaction.

3 Cost Accounting Framework

Consider the transaction in Figure 3 where a sender converts origin currency into a stablecoin at a domestic exchange a, transfers it over a blockchain network n, and the receiver converts it into destination currency at a foreign exchange b. The transfer involves two fiat currencies, the origin currency O and the destination currency D, and passes through a stablecoin S pegged to a reference currency R, giving the sandwich $O \to S \to D$. In the empirical analysis R is USD and $O \neq D$, but the framework is kept general.

We track a transfer through five stages, giving six balances indexed 0 through 5 (Table 1). The letter gives the type of asset—B fiat at a bank, X fiat at an exchange, Q stablecoin at an exchange—and the subscripts the stage and the currency. Fees follow the same logic, the letter naming the rail: f and F for fiat rail fees, τ for trading fees, and g and G for blockchain related fees, where lowercase denotes proportional and uppercase nominal fees, with subscripts giving the exchange, the payment method m or network n. In addition, time indices show how a fee or cost parameter varies: t for continuous market prices, $\bar{t}$ for administratively set fees fixed in the short run, and $\tilde{t}$ for fees that may be either. Some fees are zero in practice. Appendix Table A1 summarizes all the notation.

Table 1: The Stages and Balances of a Cross-border Stablecoin Transfer

Stage	Balance	Held at	After
0	$B_{0,O}$	Sender's bank account	—
1	$X_{1,O}$	Origin exchange a	Fiat deposit
2	$Q_{2,S}$	Origin exchange a	Stablecoin purchase
3	$Q_{3,S}$	Destination exchange b	Stablecoin transfer
4	$X_{4,D}$	Destination exchange b	Stablecoin sale
5	$B_{5,D}$	Receiver's bank account	Fiat withdrawal

3.1 Prices and Pricing Wedges

Converting fiat into a stablecoin, and back again, requires a price. Beyond the fees set by exchanges, the user therefore faces costs arising from market forces. The natural benchmark is the rate prevailing in deep global foreign currency exchange markets. Define $E_{O/R,t}$ for that rate, in units of O per unit of R, and $E_{D/R,t}$ for its destination-currency counterpart. However, neither the sender nor the receiver transacts at these rates. Instead, they transact at local crypto exchanges, where local market conditions typically give rise to two wedges, a premium and a spread.

The premium: An entire local market may be priced away from the benchmark. This is commonly referred to as the *premium*, the proportional deviation of the local mid-price

from the benchmark,

$$\rho_{a,t} \;=\; \frac{P^{\text{mid}}_{a,t}}{E_{O/R,t}} - 1, \qquad \rho_{b,t} \;=\; \frac{P^{\text{mid}}_{b,t}}{E_{D/R,t}} - 1, \tag{1}$$

where P^{mid} is the midpoint of the best bid and ask price.[6] It shifts every quote in the market together and may take either sign.[7] It plays a similar role to the exchange-rate margin in traditional remittance channels. The premium may arise for different reasons, for example, where residents face limits on obtaining foreign currency through banks, a stablecoin offers an alternative access to foreign currency, potentially resulting in prices above the benchmark. Conversely, where access is unrestricted and capital markets exhibit few frictions, the gap is close to zero.

The spread: A bid-ask spread is the difference between the ask price and the bid price for a stablecoin in the local market. The distance from the mid-price to each side of the book is the *half-spread*,

$$\delta_{a,t} \;=\; \frac{P^{\text{ask}}_{a,t}}{P^{\text{mid}}_{a,t}} - 1, \qquad \delta_{b,t} \;=\; 1 - \frac{P^{\text{bid}}_{b,t}}{P^{\text{mid}}_{b,t}}. \tag{2}$$

The bid–ask spread is the full distance between the two sides and each party crosses half of it, so it is the half-spread that enters the cost of a single transfer; we refer to δ simply as the spread below for brevity. The sender buys and pays the ask-side half at exchange a, the receiver sells and gives up the bid-side half at exchange b. The spread compensates market makers and liquidity providers for immediacy and is a cost for the user.

Together, the premia and the spread give the prices at which the transfer executes,

$$P^{\text{ask}}_{a,t} \;=\; \underbrace{E_{O/R,t}}_{\text{benchmark}}\ \underbrace{(1+\rho_{a,t})}_{\text{premium}}\underbrace{(1+\delta_{a,t})}_{\text{spread}}, \tag{3}$$

$$P^{\text{bid}}_{b,t} \;=\; \underbrace{E_{D/R,t}}_{\text{benchmark}}\ \underbrace{(1+\rho_{b,t})}_{\text{premium}}\underbrace{(1-\delta_{b,t})}_{\text{spread}}. \tag{4}$$

Spreads enter symmetrically, on both sides of the transaction, as a cost. The premia do not. Either premium may be positive or negative; what differs is how each enters the cost, because the sender buys at the origin while the receiver sells at the destination. A higher premium at the origin means the sender obtains fewer stablecoins per unit of currency,

[6] The benchmark rate prices the reference currency R while the local quote prices the stablecoin S, so ρ captures both the local premium on access to R and any deviation of the stablecoin from its peg. The two are not separately identified in our data.

[7] The premium suggests an arbitrage opportunity, but closing it requires moving fiat into and out of the local market, which capital controls, restricted bank access to foreign currency, and other frictions may impede.

raising the cost; a higher premium at the destination means the receiver obtains more currency per stablecoin, lowering it. It is therefore the difference between the two premia that matters, and total cost can be negative when the destination premium exceeds the origin premium by enough to offset all fees.

3.2 Stage 1: Fiat Deposit

The sender moves $B_{0,O}$ from a bank account to exchange a using payment method m (e.g., bank or card transfer), incurring proportional fee $f^{\text{dep}}_{a,m,\bar{t}}$ and nominal fee $F^{\text{dep}}_{a,m,\bar{t}}$. The balance credited is

$$X_{1,O} \;=\; B_{0,O}\left(1 - f^{\text{dep}}_{a,m,\bar{t}}\right) - F^{\text{dep}}_{a,m,\bar{t}}. \tag{5}$$

3.3 Stage 2: Stablecoin Purchase

The sender buys stablecoin S at exchange a at the ask price $P^{\text{ask}}_{a,t}$, and the exchange deducts a proportional trading fee $\tau_{a,\bar{t}}$ from the fiat balance before execution.[8] The stablecoin balance acquired is

$$Q_{2,S} \;=\; \frac{X_{1,O}}{E_{O/R,t}\,(1+\rho_{a,t})\,(1+\delta_{a,t})\,(1+\tau_{a,\bar{t}})}. \tag{6}$$

3.4 Stage 3: Stablecoin Transfer

The sender withdraws the stablecoins from exchange a to exchange b over network n (e.g., Ethereum or Tron), incurring withdrawal fees $g^{\text{wd}}_{a,n,\tilde{t}}$ and $G^{\text{wd}}_{a,n,\tilde{t}}$ at exchange a and deposit fees $g^{\text{dep}}_{b,n,\tilde{t}}$ and $G^{\text{dep}}_{b,n,\tilde{t}}$ at exchange b.[9] Unlike those of the first and last stages, these fees are charged in stablecoin units. The balance credited at exchange b is

$$Q_{3,S} \;=\; \left[\,Q_{2,S}\left(1 - g^{\text{wd}}_{a,n,\tilde{t}}\right) - G^{\text{wd}}_{a,n,\tilde{t}}\,\right]\left(1 - g^{\text{dep}}_{b,n,\tilde{t}}\right) - G^{\text{dep}}_{b,n,\tilde{t}}. \tag{7}$$

3.5 Stage 4: Stablecoin Sale

The receiver sells at exchange b at the bid price $P^{\text{bid}}_{b,t}$, and exchange b deducts a proportional trading fee $\tau_{b,\bar{t}}$ from the proceeds. The fiat balance credited is

$$X_{4,D} \;=\; Q_{3,S}\,E_{D/R,t}\,(1+\rho_{b,t})\,(1-\delta_{b,t})\,(1-\tau_{b,\bar{t}})\,. \tag{8}$$

[8] We adopt the budget-constraint convention when buying, so that the fiat balance must cover both the purchase price and the fee, and the haircut convention when selling in stage 4, so that the fee is deducted from fiat proceeds after execution. This follows exchange practice and the direction of the transaction. For small fees the two are approximately equivalent, since $1/(1+\tau) \approx 1-\tau$.

[9] Blockchain networks charge gas fees that exchanges typically pass on through crypto withdrawal fees, though exchanges may smooth or administratively fix them in the short run, and may apply a markup or discount. We therefore index them with $\tilde{t}$.

3.6 Stage 5: Fiat Withdrawal

The receiver withdraws $X_{4,D}$ to a bank account via payment method m, incurring proportional fee $f^{\text{wd}}_{b,m,\bar{t}}$ and nominal fee $F^{\text{wd}}_{b,m,\bar{t}}$.[10] The final balance received is

$$B_{5,D} \;=\; X_{4,D}\left(1 - f^{\text{wd}}_{b,m,\bar{t}}\right) - F^{\text{wd}}_{b,m,\bar{t}}. \tag{9}$$

3.7 The Cost of a Transfer

The five stages determine the receiver's final balance $B_{5,D}$ from the sender's initial balance $B_{0,O}$ and the fees and wedges along the way. We measure the cost as the shortfall relative to a frictionless conversion at the benchmark rate,

$$C \;=\; 1 - \frac{B_{5,D}}{B_{0,O}\,E_{D/O,t}}, \qquad E_{D/O,t} \;=\; \frac{E_{D/R,t}}{E_{O/R,t}}, \tag{10}$$

where $E_{D/O,t}$ is the implied benchmark rate between the two fiat currencies. A value of $C = 0$ is a frictionless transfer, $C > 0$ a net cost, and $C < 0$ a transfer in which the destination premium more than offsets everything else.

3.8 An Illustrative Transfer

A stylized example makes the stages concrete. A sender transfers 200 units of origin currency, facing premia of 80 basis points at the origin and 30 at the destination, spreads of 10 and trading fees of 40 basis points on each leg, fiat rail fees of 25 basis points at each end, and a flat stablecoin withdrawal fee of 0.5. The origin currency is the reference currency, so $E_{O/R,t} = 1$, and the destination currency trades at four units per unit of the reference currency. Figure 4 traces the balance through the five stages, and Table A2 in the Appendix gives every parameter and evaluates each equation.

The amount sent and the amount received are in different currencies, so the cost is not the difference between them. Valuing every balance at the benchmark rate makes the stages comparable: the transfer loses 4.95 reference-currency units of the 200 sent, which is the 2.47 percent given by equation (10).

3.9 From Routes to Corridors

Equation (10) gives the cost of a transfer given the choice of exchanges, stablecoin, and network. A sender faces many such combinations. Thus, in a single corridor, there are many routes for the transaction to take. We therefore distinguish between rails, routes, and corridors.

[10] Payment method m in Stage 5 is not necessarily the same as payment method m in Stage 1.

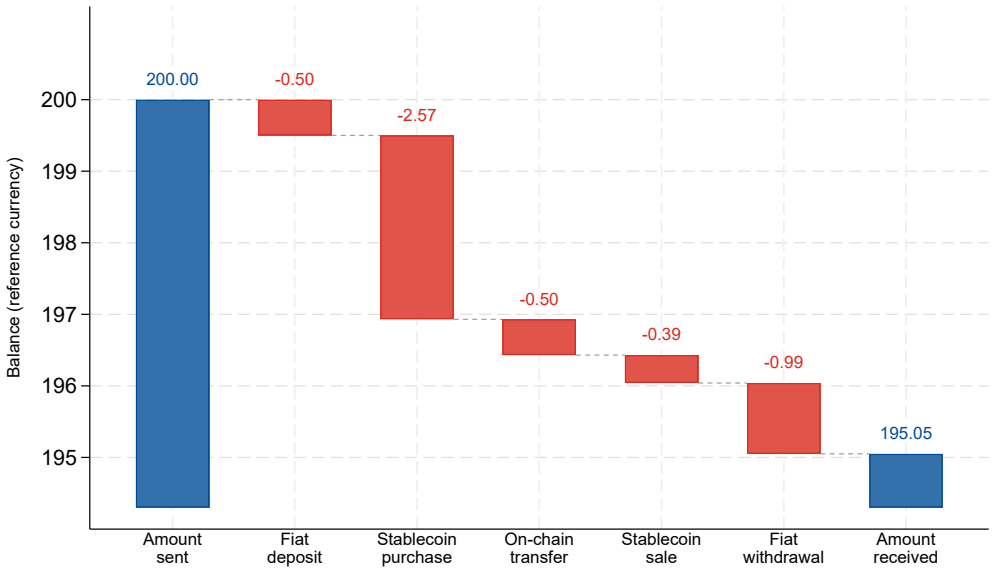


Figure 4: The Balance Through the Five Stages

Notes: Balances are converted into reference-currency units at the benchmark rates, following the final column of Table A2. Each intermediate bar shows the amount lost at that stage.
Source: Authors' calculations.

A *rail* is the point at which a user enters or leaves the stablecoin system: a country, exchange, stablecoin, and blockchain network, such as Brazil–Binance–USDT–Tron. A *route* is a pair of rails, one at each end, matched on the stablecoin and the network. Thus, by assumption, the sender cannot send USDC over Ethereum and have the receiver take delivery of USDT over Tron. Define a route as $r = (a, b, S, n)$, with cost C_r. Routes are directional: Brazil–Binance–USDT–Tron to USA–Coinbase–USDT–Tron and the reverse are distinct routes that may have different costs. Fiat deposit and withdrawal methods m do not create routes, since the choice of method does not change which rails connect, but one must be observed at each end for a route to be priced.

Not every rail forms a route, and not every route can be priced. A rail connects only to rails carrying the same coin on the same network, so one with no counterpart at the other end forms no route. A route is *potential* when both its rails are active, and *complete* when fees, market prices, and weights required for are observed. Define $\mathcal{R}_{OD}$ as the set of complete routes in a corridor.

A *corridor* is a directed pair of countries $O \to D$, and a corridor cost is a weighted average of its route costs, where the weights are given by how much each route is used. Route-level usage is not directly observed, but we observe the four marginals and assume the joint distribution is their product, so the weight before normalization on route r is

$$\omega_r = \underbrace{s^{\text{exch}}_{a|S,O}}_{\text{origin exchange}} \times \underbrace{s^{\text{exch}}_{b|S,D}}_{\text{destination exchange}} \times \underbrace{s^{\text{coin}}_{S}}_{\text{coin}} \times \underbrace{s^{\text{chain}}_{n|S}}_{\text{network}}. \tag{11}$$

Each s is a share, and the conditioning bar states what it is a share of. The exchange shares are local, where for example, $s^{\text{exch}}_{a|S,O}$ is exchange a's share of trade in stablecoin S against the origin currency, across the exchanges observed in country O. The coin and network shares are global measurements, where network shares are conditional on coin.

Weights are normalized over the routes that can be priced,

$$w_r \;=\; \frac{\omega_r}{\sum_{r' \in \mathcal{R}_{OD}} \omega_{r'}}, \tag{12}$$

where r' runs over the complete routes of the same corridor, $\mathcal{R}_{OD}$. The corridor cost is then a weighted average of all the complete routes

$$C_{OD} \;=\; \sum_{r \in \mathcal{R}_{OD}} w_r \, C_r. \tag{13}$$

Two features of equation (12) are worth noting. The denominator runs over $\mathcal{R}_{OD}$ rather than over all routes connecting the two countries, so w_r is route r's share of usage among the routes we can price; a route missing data has its weight redistributed proportionally over those that remain. Where only a single exchange, coin, or chain has a complete route, that option receives full weight regardless of its global share.

4 Data Collection and Estimation

Country and exchange selection. We select countries in our sample from among the top 40 remittance-sending and top 40 remittance-receiving economies, ranked by gross flows using the latest WDI data (World Bank, 2024). The sample spans a diverse set of income groups and regions, and ensures substantial overlap (exceeding 95 percent) with the Remittance Prices Worldwide (RPW) dataset, our benchmark on the cost of cross-border transfers through traditional channels.[11]

For each country, we select crypto exchanges based on the size of stablecoin trades, using data from CoinDesk (formerly CryptoCompare).[12] We focus on centralized exchanges that trade USDT and USDC for local currency and we match trades involving a given local currency to the country that issues it (e.g. Mexico for stablecoin trades in Mexican Pesos), regardless of where the exchange is domiciled and licensed to operate. Trades in global crypto exchanges such as Binance or OKX are allocated to specific countries based on the currency used. We exclude dollarized economies from the sample and aggregate data for euro area countries, which cannot be distinguished by a unique currency. We

[11] While corridor-level remittance flow data would have been a preferred choice, the World Bank KNOMAD data does not reflect directly observed flows and has been discontinued after 2021.

[12] CoinDesk offers rich data on the price of tokens and trade volumes and has been used in several economics papers such as Aldasoro et al. (2026).

then compute exchange market shares based on total observed stablecoin trading volumes and values for 2025.[13] For each country, we select a sufficient number of exchanges so that we cover at least 80% of the trading volumes and 80% of the trading values in each local currency-stablecoin pair.[14] Exchanges for which fee data is not available drop from the sample. After applying these selection criteria, and accounting for data availability, our final sample comprises 18 countries and 29 distinct exchanges.[15]

Blockchain selection. Each stablecoin operates on many blockchains. While USDC and USDT both began as ERC-20 tokens on the Ethereum blockchain, they can be used across a range of blockchains.[16] Exchanges now offer deposits and withdrawals across a number of blockchains. The CoinDesk data does not indicate which blockchain networks each exchange supports, so we focus on the most frequently used chains globally. For USDC, we include Ethereum, Solana, and Base, which together accounted for 90% of the total volumes of on-chain activity in 2025. For USDT, we include Tron, Ethereum, and Binance Smart Chain, which accounted for more than 90% of on-chain volumes. We use data from Allium to determine these most used chains. We note that, while the stablecoins are on a much wider set of chains, we capture a large portion of the volumes, while covering chains that are offered by many of the exchanges from which we collect fee data.

Transfer size. In the baseline scenario, we report fees applicable to a $200 transfer. This corresponds to the standard benchmark transaction size in the RPW data and is broadly representative of a typical low-value migrant remittance. Consistent with the RPW methodology, this amount is expressed in nominal terms to facilitate cross-country comparisons and is not adjusted for purchasing power parity.

Exchange fees. For each exchange in our sample, we collect fiat rail, trading, and crypto transfer fees from the exchanges' websites, within a short time window at the beginning of 2026 Q2. We rely exclusively on information disclosed publicly by the exchanges themselves. We do not report fees that are revealed only after authentication, as they may vary across users, nor do we include information from third-party aggregator websites. When necessary, we use a local IP address to access country-specific fee schedules. In the baseline scenario, we report fees applicable to a $200 transfer. In exchanges

[13] We only keep exchanges with trades in a currency-stablecoin over the course of 2025. In some cases, exchanges offer stablecoins at par directly and not through a market mechanism. We do not have data on the volumes of such trades and therefore do not include them.

[14] In practice, this approach covers over 90 percent of market shares for USDT trades and 99 percent of market shares for USDC trades in our sample.

[15] Because several exchanges operate in multiple countries, this corresponds to 55 country-exchange observations.

[16] ERC-20 is a technical specification of the implementation of the tokens on the Ethereum blockchain.

that implement tiered pricing, a $200 transfer always corresponds to an entry-level tier in our sample. We assume that the sender and receiver do not benefit from preferential fee schedules, such as membership discounts or temporary promotions.

We note the following:

- Fiat deposit and withdrawal fees: We collect data for all available payment methods (ACH, wire, card, instant payments, digital wallets, etc). Importantly, we report only fees charged by the exchange itself. In some cases, deposit contractual agreements with banks may allow for additional transfer fees that may vary across users and that we are not able to observe. For fiat deposit and withdrawal methods, we collect all available options but select the cheapest available method at each exchange as the baseline. In practice, this most commonly corresponds to standard bank transfers, which in most countries are a convenient and widely used means of transferring funds. We assess the sensitivity of this choice in Section 6.

- Trading fees: Exchanges typically apply differentiated maker and taker fees. We collect the taker fee—corresponding to a market order that executes immediately—which is typically higher than a maker fee.

- Crypto transfer fees, for a given token and chain, adjust frequently to reflect network congestion. We observe that for a given token and chain, many international exchanges in our sample adopt dynamic pricing models, updating crypto withdrawal fees in real time. To ensure comparability across exchanges, we collect crypto transfer fee data from global exchanges within a one-hour window. In addition, global exchanges typically do not charge crypto transfer fees for cross-border transactions that remain within the same platform (e.g. Kraken USA to Kraken Canada). Accordingly, we set crypto transfer fees to zero where the origin and destination exchanges are the same.

Spreads and premiums. In addition to the exchange fees, users face a bid-ask spread and a local price premium when trading on exchanges. The bid-ask spread is not directly observed in the CoinDesk data we use, since we only consider a daily price and volume for each exchange-token-fiat observation. We instead estimate the bid-ask spread from an algorithm in the traditional finance literature (Corwin and Schultz, 2012), which relies on observing the low and high for each day. We use the daily high and low from each exchange-coin pair over 2025 to compute a daily spread. We then take a weighted median for each exchange over 2025. We find that the spread is a relatively small part of the costs.[17]

[17]The algorithm uses the assumption that intraday prices follows Brownian motion and estimates the bid and ask spread for every day. We use a version of the algorithm that does not impose non-negativity,

We define the premium as the percentage deviation of the daily price of token in local currency from a benchmark exchange rate. The local premium at exchange a is defined as

$$\rho_{a,t} \;=\; \frac{P^{\text{mid}}_{a,t}}{E_{O/R,t}} - 1.$$

We take the benchmark exchange rate levels $E_{O/R,t}$ from Haver Datastream.[18] We calculate $P^{\text{mid}}_{a,t}$ as the midpoint of the high and low for each day for each exchange and coin-fiat pair. For the United States, we use $E_{O/R,t} = 1$. To aggregate for calculation of overall costs we take the percentiles (25th 50th and 75th) of the premium weighting by total daily number of trades on that exchange. We exclude days with premiums larger than 20% in absolute value from the calculation of these percentiles as we want to focus on unsophisticated households using stablecoins for transfers rather than arbitrageurs.[19]

Summary statistics. Table 2 presents summary statistics for the different types of fees and costs collected for a $200 transfer. Panel A indicates that trading fees average 0.37 percent, with a median of 0.20 percent. This reflects that many exchanges impose low or zero maker fees, whereas taker fees can reach up to 1.20 percent. The premium averages 0.41 percent, with a standard deviation of 1.01 percent and a range of -2.04 to 4.81 percent, demonstrating substantial variation across exchange-currency pairs. This is broadly consistent with Aldasoro et al. (2026). The bid-ask spread is more contained, averaging 0.20 percent. Panel B shows that crypto withdrawal fees are predominantly nominal rather than proportional; the median percentage fee is zero, while the median nominal fee is $0.80. The table further indicates that crypto deposit fees are essentially non-existent. Fiat rail fees, as shown in Panels C and D, are highly dispersed and right-skewed. Most exchanges offer at least one free or low-cost deposit method, resulting in median fees of zero, while the upper end ranges from 4.50 percent for deposits to 3.50 percent for withdrawals. Panel C presents the full set of fiat rail methods available, whereas Panel D restricts the analysis to the cheapest option per exchange.

Construction of Routes and Corridor-Level Cost Estimates. Section 3.9 defines rails, routes, and corridors. The sample comprises 258 rails across 18 countries and 29 distinct exchanges, operating over 55 country-exchange pairs, with an average of 14.3 rails per country (Table 3, Panel A).

and as a result some of the daily observations have a negative bid-ask spread. We present the weighted by number of trades median for each exchange-token-fiat observation across the days the token was traded within 2025. None of the median estimates are negative.

[18] Where possible we use data from the Bank of International Settlements. A complete list of the exchange rates used is available upon request.

[19] In results beyond the scope of this paper, it appears that most of the variation in the premium for a given coin-fiat pair is in the time dimension rather than across exchanges. We preserve the variation across exchanges as we seek to capture cross-exchange rather than cross-time variation.

Table 2: Fee Schedule and Market Price Summary

	Mean	SD	Min	Median	Max	N
Panel A. Trading fees and market prices						
Trading fee, %	0.37	0.36	0.00	0.20	1.20	103
Premium, %	0.41	1.01	−2.04	0.06	4.81	89
Spread, %	0.20	0.20	0.00	0.15	1.14	89
Panel B. Crypto transfer fees						
Withdrawal fee, %	0.02	0.12	0.00	0.00	1.00	201
Withdrawal fee, nominal	1.09	2.02	0.00	0.80	20.00	199
Deposit fee, %	0.01	0.12	0.00	0.00	1.00	215
Deposit fee, nominal	0.02	0.14	0.00	0.00	1.00	215
Panel C. Fiat rail fees — all observed methods						
Deposit fee, %	0.73	1.29	0.00	0.00	4.50	93
Deposit fee, nominal	0.66	2.96	0.00	0.00	21.40	93
Withdrawal fee, %	0.30	0.79	0.00	0.00	3.50	74
Withdrawal fee, nominal	1.40	4.30	0.00	0.00	25.00	74
Panel D. Fiat rail fees — cheapest method						
Deposit fee, %	0.19	0.72	0.00	0.00	4.50	49
Deposit fee, nominal	0.03	0.17	0.00	0.00	1.16	49
Withdrawal fee, %	0.20	0.65	0.00	0.00	3.00	45
Withdrawal fee, nominal	0.34	0.87	0.00	0.00	5.00	45

Notes: All values are expressed as a percentage of the transaction amount, except nominal fees which are in US dollars. Panel A reports trading fees and market prices at the (country, exchange, stablecoin) level. Trading fees are percentage fees charged on each transaction. The premium is the percentage deviation of the local stablecoin price from the benchmark fiat exchange rate; a negative value indicates the stablecoin trades below the benchmark. The spread is the half bid-ask spread. Panel B reports crypto transfer fees at the (exchange, stablecoin, blockchain) level, separately for withdrawal and deposit. Panels C and D report fiat deposit and withdrawal fees across all observed payment methods per exchange (Panel C) and for the cheapest method only (Panel D). N is the number of observations with non-missing values for each variable.
Source: Authors' calculations.

As shown in Panel B, the sample includes 10,077 potential routes and 4,005 complete routes, with an average of 32.9 potential and 13.1 complete routes per corridor. Some corridors have as many as 180 routes, while others have no potential routes at all. The baseline sample includes 270 corridors out of 304 potential directed pairs, averaging 15 complete corridors per origin country (Panel C). The remaining 34 pairs lack available routes or sufficient data to form a complete route.

For each complete route we compute the transfer cost C_r by applying the framework of Section 3 to the fees and market prices observed along the five stages of that route.

The route weights of equation (12) draw on three sources. Exchange shares are computed from trade counts at the (country, exchange, stablecoin) level; a volume-based alternative is reported in Section 6. Coin weights are the global on-chain market shares of each stablecoin, 58.25 percent for USDT and 41.75 percent for USDC. Chain weights

conditional on the coin are, for USDT, Tron 58.32 percent, Ethereum 26.26 percent, and BSC 15.42 percent; and for USDC, Ethereum 73.35 percent, Solana 17.20 percent, and Base 9.45 percent. Fiat deposit and withdrawal methods do not enter the weighting scheme, since no equivalent usage data exist to determine which methods users select; the cheapest available method at each exchange is used as the baseline, as discussed above.

Table 3: Sample Coverage by Country, Route, and Corridor

	Mean	SD	Min	Max	Total
Panel A. Coverage per country					
Exchanges per country	3.1	1.9	1	8	55
Rails per country	14.3	8.8	3	33	258
Panel B. Route coverage per corridor					
Potential routes per corridor	32.9	30.2	0	180	10,077
Complete routes per corridor	13.1	12.4	0	67	4,005
Panel C. Corridor coverage per country					
Potential corridors, as origin	16.9	0.3	16	17	304
Potential corridors, as destination	16.9	0.3	16	17	304
Complete corridors, as origin	15.0	5.3	0	17	270
Complete corridors, as destination	15.0	0.6	13	16	270

Notes: The unit of observation in Panel A is the country. A rail is a (country, exchange, stablecoin, blockchain) combination. Panel B reports route coverage across the 306 directed country-pair corridors in the sample. A potential route is a directed pair of rails matched on stablecoin and blockchain where both rails are active. A complete route is a potential route for which all fee, market price, and weight data required to estimate the transfer cost are observed. Panel C reports corridor coverage from the perspective of each country as origin and as destination. Statistics are computed across the 18 sample countries and 29 distinct exchanges.
Source: Authors' calculations.

5 Results

Table 4 presents summary statistics for corridor-level transfer costs. The mean corridor cost is 2.18 percent of a $200 transfer, while the median is 1.99 percent, indicating a modest right-skew in the distribution.[20] Corridor costs vary from −2.18 percent to 8.03 percent across the 270 corridors in the sample, demonstrating substantial heterogeneity. Panel C of Table 4 indicates that, at the route level prior to corridor aggregation, costs are even more dispersed, ranging from −4.15 to 10.71 percent across 4,005 complete routes, with a mean of 2.09 percent, consistent with the corridor-level estimate.

Panel B of Table 4 presents a decomposition of corridor costs into four components.[21] Trading fees are the largest component, averaging 0.85 percent, followed by fiat rail fees

[20] Each corridor enters the mean, median, min, and max with equal weight, so these statistics describe the corridors we observe rather than an average weighted by the size of flows.

[21] Because the wedges of Section 3 are proportional, they interact multiplicatively, and we approximating the loss from a proportional wedge f applied to a balance Z as Zf, while a nominal fee contributes its full amount. The approximation is close because the individual wedges are small: at a premium of 50

at 0.67 percent and crypto transfer costs at 0.48 percent. The premium and spread component averages 0.18 percent but displays the greatest variability, with a standard deviation of 1.33 percent and a range from −3.07 to 4.51 percent. The negative minimum indicates corridors where the destination-side premium is sufficiently large to offset all other cost components, which aligns with the negative total costs reported in Panel A. Collectively, fiat rail fees and trading fees comprise the majority of average transfer costs, whereas the crypto transfer component, often identified as the primary source of cost savings in stablecoin-based payments, accounts for less than one quarter of total costs on average.

Table 4: Corridor Cost Estimate and Decomposition for a $200 transfer

	Mean	SD	Min	Median	Max	N
Panel A. Corridor cost						
Total cost, %	2.18	1.84	−2.18	1.99	8.03	270
Panel B. Corridor cost decomposition						
Fiat rails, %	0.67	1.04	0.00	0.31	4.86	270
Trading fees, %	0.85	0.50	0.09	0.88	1.98	270
Crypto transfer, %	0.48	0.38	0.00	0.45	2.40	270
Premium and spread, %	0.18	1.33	−3.07	0.11	4.51	270
Panel C. Route cost						
Total cost, %	2.09	1.87	−4.15	1.82	10.71	4,005

Notes: The unit of observation in Panels A and B is the corridor (N = 270 directed country pairs). Corridor costs are weighted averages across complete routes, using exchange weights, coin weights, and chain weights conditional on coin. The decomposition in Panel B follows the four-group classification: fiat rails (fiat deposit and withdrawal fees), trading fees (exchange trading fees on both legs), crypto transfer (crypto withdrawal and deposit fees), and premium and spread (market price deviation and bid-ask spread on both legs). Panel C reports statistics at the route level (N = 4,005 complete routes) without weighting.
Source: Authors' calculations.

Table 5 provides a more detailed breakdown of costs into ten distinct fee and pricing components across the five stages of the payment chain. On the origin side, trading fees (0.45 percent), the premium (0.44 percent), and crypto withdrawal fees (0.47 percent) are of similar magnitude, followed by fiat deposit fees at 0.3 percent. The spread is notably smaller at 0.11 percent, which reflects the relatively tight bid-ask spreads observed within the sample. In contrast, the destination-side cost structure differs significantly. Crypto deposit fees (0.01 percent) are essentially nonexistent, and the spread (0.11 percent) is again minimal, whereas trading fees (0.41 percent) and fiat withdrawal fees (0.37 percent) constitute the majority of destination-side costs. The destination-side premium averages −0.48 percent. When stablecoins trade at a premium to the official exchange rate, the

basis points, a spread of 10, and a trading fee of 40, the linear sum is roughly 100 basis points while the interaction terms come to less than 0.3. Grouping by type gives the four categories reported here—f and F for fiat rail fees, τ for trading fees, g and G for blockchain transfer costs, and ρ and δ for market pricing frictions.

seller receives more destination currency per unit of stablecoin, thereby reducing the overall transfer cost. In a balanced panel, origin and destination premiums would largely offset each other. In our estimates of the average corridor, the net premium contribution is close to zero at $0.44 - 0.48 = -0.04$ percent, but does not cancel exactly because the sample of 270 directed corridors is unbalanced; not all country pairs appear in both directions.

Table 5: Detailed Corridor Cost Decomposition for a $200 transfer

	Mean	SD	Min	Median	Max	N
Panel A. Origin side						
Fiat deposit, %	0.30	0.87	0.00	0.00	3.66	270
Trading fee, %	0.45	0.36	0.00	0.29	0.99	270
Premium, %	0.44	0.93	−0.90	0.10	3.80	270
Spread, %	0.11	0.11	0.00	0.07	0.55	270
Crypto withdrawal, %	0.47	0.38	0.00	0.45	2.40	270
Panel B. Destination side						
Crypto deposit, %	0.01	0.03	0.00	0.00	0.13	270
Trading fee, %	0.41	0.35	0.00	0.25	1.00	270
Premium, %	−0.48	0.89	−2.96	−0.12	0.81	270
Spread, %	0.11	0.11	0.00	0.07	0.56	270
Fiat withdrawal, %	0.37	0.56	0.00	0.30	2.48	270

Notes: The unit of observation is the corridor (N = 270). All values are expressed as a percentage of the transfer amount. Panel A covers the five cost components incurred at the origin exchange, in waterfall order: the sender pays a fiat deposit fee, incurs trading fees and the stablecoin premium and spread upon purchase, and pays the crypto withdrawal fee. Panel B covers the five components at the destination exchange: the receiver incurs a crypto deposit fee, trading fees, the stablecoin premium and spread upon sale, and a fiat withdrawal fee. Premium is the deviation of the stablecoin price from the official exchange rate; a negative value indicates that stablecoins trade at a premium to the official rate at that leg.
Source: Authors' calculations.

Table 6 presents a comparison of costs across corridor groups. Specifically, we compare the 15 corridors with the lowest costs with the 15 corridors with the highest costs. The 15 lowest-cost corridors exhibit a mean total cost of −1.05 percent. All four cost components are below their respective sample averages (Panel B), with the premium and spread component serving as the primary driver, averaging −2 percent. In contrast, the 15 highest-cost corridors average 6.88 percent, primarily due to two components: fiat rail fees and the premium and spread. The difference in the premium and spread component between the three groups is almost entirely driven by the premium, while spreads are largely comparable across all corridor groups. Overall, this decomposition suggests that market pricing frictions—in particular the premium—and domestic payment infrastructure, rather than the aspects of stablecoin trade and transfers, are the primary determinants of cross-corridor cost heterogeneity.

Table 6: Cost of a $200 transfer, by Corridor Group

	All corridors	Lowest-cost	Highest-cost
Panel A. Total corridor cost			
Total cost, %	2.18	−1.05	6.88
Panel B. Cost decomposition			
Fiat rails, %	0.67	0.04	3.28
Trading fees, %	0.85	0.72	0.68
Crypto transfer, %	0.48	0.19	0.42
Premium and spread, %	0.18	−2.00	2.50
N corridors	270	15	15

Notes: Each row reports the simple mean across the indicated group of corridors. The unit of observation is the corridor. The lowest- and highest-cost corridor groups are selected by total cost; the number of corridors in each group is shown in the bottom row. The decomposition follows the four-group classification described in Table 4. All values are expressed as a percentage of the transfer amount.
Source: Authors' calculations.

5.1 Corridor Heterogeneity

The 270 corridors in the sample exhibit substantial variation in transfer costs, and Table 6 demonstrates heterogeneity in the underlying cost components. Figure A1 shows the distribution of corridor costs across the sample. The distribution is unimodal and right-skewed, with a long tail of high-cost corridors that raises the mean above the median. Roughly a quarter of corridors have costs below 1 percent, while a small subset displays negative costs. In contrast, a few corridors have costs exceeding 6 percent, well above the G20 target of 3 percent.

Figures A2, A3, and A4 present some more detailed information by corridor. Costs are highly asymmetric; the cost of sending from one country to another often differs substantially from the reverse direction. These differences might be due to each of the cost components considered, in both the origin and destination country, underscoring the importance of local market conditions. Figure A5 illustrates within-corridor heterogeneity—for a given corridor, individual routes vary considerably in cost, offering both low- and high-cost alternatives to users.

We also compare corridors by the development classification of the sending and receiving country, distinguishing advanced economies (AEs) from emerging market and developing economies (EMDEs). Average corridor costs differ across the four directed groups: AE→AE corridors average 2.31 percent, AE→EMDE 1.61 percent, EMDE→AE 3.05 percent, and EMDE→EMDE 2.01 percent (Table A7, Panel A).

The premium accounts for much of this pattern. Because senders buy stablecoins in the origin market and sell them in the destination market, a positive premium at the destination reduces cost while a positive premium at the origin raises it. Since premia are typically positive in EMDE markets, sending to an EMDE is cheaper and sending from

one is more expensive (Figure A7). Setting the premium to zero confirms this: average costs move in opposite directions for the two groups: AE→EMDE rises to 2.00 percent and EMDE→AE falls to 2.72 percent, while the two same-class groups are essentially unchanged on average, as origin and destination premia offset (Table A7, Panel B). Corridors that send from an EMDE are considerably more dispersed than those sending from an AE (Figures A8 and A9), and removing the premium roughly halves the standard deviation within every group and eliminates all negative-cost corridors (Table A7, Panel B).

The fiat rails also play a role. Fiat costs are origin-driven as EMDE fiat deposit costs are higher on average, while withdrawal costs are effectively equal across AE and EMDE destinations. Trading fees are higher on average on AE exchanges, and crypto transfer costs are similar across groups (Figure A7).

5.2 Comparison with Traditional Routes

We compare our results with the cost of transfers via traditional remittance routes, using data from RPW. Table 7 reports summary statistics for a $200 transfer based on averages within each corridor. The RPW data covers four main types of remittance service providers, namely banks, money transfer operators, mobile operators, and post offices. Under the RPW methodology, transfer fees are recorded only at the initiation point and therefore represent a lower bound on total fees.[22] Even so, traditional routes appear substantially more costly than stablecoin transactions, with an average cost of 7.18 percent and a median cost of 5.14 percent in the full RPW sample.[23] Costs along traditional routes also exhibit greater dispersion at the corridor level, with outliers reaching up to 118.19 percent.

These findings are robust when the sample is restricted to the same corridors, ruling out a country sample selection bias. Traditional routes are more expensive than stablecoin routes, with an average cost of 5.81 percent and a median of 4.96 percent, almost 5 percentage points above the cost of stablecoins. In addition, the result that stablecoins are cheaper on average holds—both in the unrestricted and restricted samples—when considering the cheapest available route within each corridor. Among traditional routes, it is noteworthy that transactions initiated via the internet are less expensive in baseline scenarios but appear more costly among the cheapest routes.

Figure A6 shows pairwise correlations between stablecoin and traditional route costs. In the baseline scenario using average costs, stablecoins are consistently cheaper (Panel A). However, when focusing on the cheapest routes, although most observations cluster around

[22]RPW cost data comprises two components: a transaction fee and an exchange rate spread. As in our methodology, the exchange rate spread can be negative. For details on the RPW methodology, see `https://remittanceprices.worldbank.org/methodology`

[23]By comparison, the average cost of traditional transfers across all payment routes—without first averaging at the country level—is 6.36 percent as of 2025 Q3.

the 45°-line, approximately one-third of corridors exhibit lower costs through traditional routes than through stablecoins (Panel B). This result should nevertheless be interpreted with caution, as it reflects a comparison of tail outcomes across the two distributions.

Table 7: Comparison with RPW Data – $200 transfer

	Mean	SD	Min	Median	Max	N
Panel A. All corridors						
Baseline						
Stablecoin	2.18	1.84	−2.18	1.99	8.03	270
RPW, all methods	7.18	8.89	0.24	5.14	118.19	348
RPW, internet	5.86	8.97	0.20	4.26	108.30	321
Cheapest option						
Stablecoin	1.07	1.90	−4.15	0.96	6.34	270
RPW, all methods	2.12	3.27	−9.85	1.77	45.13	348
RPW, internet	2.61	6.77	−9.85	1.84	82.60	321
Panel B. Matched corridors						
Baseline						
Stablecoin	1.02	1.11	−1.64	1.23	2.97	33
RPW, all methods	5.81	3.50	1.91	4.96	16.39	33
RPW, internet	4.24	1.81	1.77	3.97	9.71	33
Cheapest option						
Stablecoin	0.04	1.45	−3.45	0.38	2.49	33
RPW, all methods	0.70	1.60	−2.65	0.75	5.29	33
RPW, internet	0.74	1.63	−2.65	0.75	5.29	33

Notes: All values expressed as a percentage of the transfer amount. The unit of observation is the corridor. Mean, SD, minimum (Min), median, and maximum (Max) refer to the distribution of corridor-level costs across corridors. Stablecoin baseline costs are weighted averages across complete routes within each corridor, using exchange trade-count weights, coin weights, and chain weights, consistent with the main results. Stablecoin cheapest option is the minimum route cost within each corridor. RPW baseline costs are computed by first averaging costs across providers within each corridor, using simple averages. RPW cheapest option is the minimum provider cost within each corridor. RPW, internet restricts to internet-based access points; RPW, all methods includes all access points including cash agents and call centers. Panel A reports the full stablecoin baseline sample (N=270) and the full RPW dataset without Euro Area aggregation. Panel B restricts to the 33 corridors present in both samples; Euro Area member states are treated as a single origin or destination in the stablecoin sample and RPW observations from individual Euro Area countries are averaged accordingly.
Source: Authors' calculations; World Bank Remittance Prices Worldwide (2025 Q3).

6 Sensitivity Analysis

Fiat payment method. Table A3 presents a sensitivity analysis of the baseline results with respect to the choice of fiat payment method. The baseline uses the lowest-cost fiat deposit and withdrawal method available at each exchange, establishing a lower bound on the fiat component. To evaluate the impact of less favorable fiat options, two additional scenarios are considered: the median-cost method and the highest-cost method available

at each exchange. A fourth scenario, termed "cheapest imputed," expands the baseline sample by imputing missing fiat fees from peer exchanges within the same country, thereby increasing the number of complete routes from 4,005 to 4,028.

Panel A of Table A3 displays the distribution of the total costs of the corridor for each scenario. The baseline mean of 2.18 percent increases to 3.04 percent under the median fiat method and to 4.07 percent under the most expensive method, illustrating the variation in fiat fees across payment methods. The cheapest imputed scenario yields results that are nearly identical to the baseline, indicating that the imputed corridors exhibit similar cost profiles. Panel B isolates the fiat rail component. Specifically, the fiat rail mean increases from 0.67 percent under the cheapest method to 1.54 percent under the median and to 2.57 percent under the most expensive method.

Weighting schemes. The baseline results apply exchange trade-count weights alongside coin and chain weights. Table A4 presents results for three alternative route weighting schemes. The three alternative approaches include volume-based exchange weightsand equal weighting across exchanges within each corridor while retaining coin and chain weights, and fully equal weighting across all exchange, coin, and chain dimensions.[24]

The mean corridor cost remains stable across all four specifications, ranging from 2.18 percent under the baseline to 2.33 percent under volume-based and equal exchange weighting. The median and minimum values also exhibit stability, with differences of no more than 0.2 percentage points across variants. The maximum cost varies to a greater extent, from 7.35 to 8.77 percent, indicating that extreme corridors are more sensitive to the weighting of individual exchanges.

Exchange trade counts and volumes proxy how stablecoin transfers are actually used, but they are themselves equilibrium outcomes shaped by the costs we measure, as well as by liquidity, premia and spreads, counterparty trust, settlement speed, and the regulatory treatment of exchanges in each jurisdiction. The usage-weighted estimates should therefore be read as the cost implied by current revealed usage patterns, not as the cost of a representative feasible route. The equal-weighted variants provide the complementary benchmark: by treating every available exchange, coin, and chain symmetrically, they approximate the cost of a route drawn at random from the feasible set.

Transfer size. Table A5 shows the variation in corridor costs as transfer amounts increase from \$50 to \$5,000. The mean corridor cost declines from 4.12 percent at \$50 to 1.54 percent at \$5,000. This downward trend results from nominal fees, which are fixed charges that represent a larger proportion of smaller transfers. Many fiat methods,

[24]We note that, in the terminology of payments, the term volume is used slightly differently than here. Here, volume means the number of stablecoins transacted on a given exchange on a given day, this corresponds more closely to the term payments value. We use the terminology of the data provider to aid replication.

and especially cryptocurrency withdrawal fees, are structured as fixed nominal fees. As indicated in Table 2, the average crypto withdrawal fee has a nominal 1 USDT/USDC component. Therefore, fixed fees raise the relative cost of smaller transfers, whereas larger transfers are less affected. However, our results also indicate that fixed fees play a larger role in traditional remittances. A $500 stablecoin transfer costs 1.79 percent on average, compared to 4.61 percent for traditional channels. The difference is narrower than for a $200 transfer.

Premium estimate. We also examine the sensitivity of our results to the choice of premium estimate. The baseline uses the median premium for each exchange-stablecoin pair. Table A6 reports results under three alternative scenarios. Setting the premium to zero—equivalent to pricing both legs of the transaction at the benchmark exchange rate—leaves the mean corridor cost nearly unchanged at 2.21 percent but compresses the standard deviation from 1.84 to 1.05 and eliminates all negative-cost corridors, confirming that the premium is the primary driver of cross-corridor heterogeneity rather than average cost levels. Similarly, using the 25th and 75th percentile premium estimates leaves the mean corridor cost largely unaffected, but shifts the distribution, with the 75th percentile producing a standard deviation of 2.46 percent and a range from -3.35 to 9.25 percent, illustrating that the cost in a given corridor is sensitive to local stablecoin pricing conditions at the time of the transfer.

7 Conclusion

This paper develops a cost accounting framework for cross-border stablecoin transfers and provides the first systematic measurement of end-to-end transfer costs across a broad set of corridors. Using 4,005 complete transfer routes across 270 directed corridors, 18 countries, 29 exchanges, two stablecoins, and five blockchain networks, we decompose the "stablecoin sandwich" into individual stages and identify which components account for the level and cross-corridor variation in transfer costs. Our results show that costs are driven primarily by the fiat access and exchange layers—including fiat deposit and withdrawal fees, trading fees, and stablecoin premiums—rather than by the blockchain transfer that connects sender and receiver.

Three findings stand out. First, the mean cost of a $200 stablecoin transfer across 270 corridors is 2.18 percent, below the G20 target of 3 percent and below the global average cost of traditional channels. However, costs vary widely, and a non-trivial share of corridors remains above the target. Second, trading fees and fiat rail fees together account for the majority of average costs, while blockchain transfer fees—the component most commonly associated with stablecoin cost savings—contribute less than one quarter. Third, the stablecoin premium is the dominant source of cross-corridor heterogeneity.

Countries where stablecoins trade at a large premium to benchmark exchange rates are expensive to send from but cheap to receive into, generating sharp asymmetries across corridor directions.

These findings carry implications for how stablecoin-based transfers are assessed. Our decomposition suggests that the cost-relevant frictions lie predominantly in the exchange and fiat access layers rather than in the blockchain infrastructure itself. More broadly, evaluations of stablecoin-based payments that focus narrowly on blockchain fees risk overstating the cost advantage. A complete assessment requires measuring the full cost stack, including the domestic legs of the transaction. In addition, our findings suggest that underlying legal and regulatory frameworks, including capital flow regimes, AML/CFT requirements, and exchange licensing regimes, together with payment infrastructure and the degree of competition, play an important role in explaining differences in transfer costs. An investigation of these drivers is left for future research.

Several limitations should be noted. Our analysis focuses on centralized-exchange-to-centralized-exchange transfers and therefore does not capture alternative transfer paths. It also focuses on observable transaction costs and does not capture other dimensions relevant to users, such as speed, non-price frictions, and counterparty risk, or regulatory and compliance costs borne by intermediaries. Our cost estimates reflect a snapshot of fees and market conditions; as the stablecoin ecosystem evolves, the cost structure may shift, including as traditional payment and remittance providers increasingly integrate stablecoin technology into their services. Extending the framework and data collection method to incorporate these dimensions and to track costs over time is a natural direction for future work.

A Appendix

Table A1: Notation

Symbol	Description	Denomination
Panel A. Currencies, exchanges, and networks		
a, b	Origin exchange, destination exchange	—
O, D	Origin currency, destination currency	—
S	Stablecoin	—
R	Reference currency to which S is pegged	—
n	Blockchain network	—
m	Payment method for a fiat deposit or withdrawal	—
Panel B. Balances		
$B_{0,O}$	Amount leaving the sender's bank account	Origin currency
$X_{1,O}$	Fiat credited at the origin exchange	Origin currency
$Q_{2,S}$	Stablecoin acquired at the origin exchange	Stablecoin
$Q_{3,S}$	Stablecoin credited at the destination exchange	Stablecoin
$X_{4,D}$	Fiat credited at the destination exchange	Destination currency
$B_{5,D}$	Amount arriving in the receiver's bank account	Destination currency
Panel C. Prices and pricing wedges		
$E_{O/R,t}$	Benchmark rate, origin currency per unit of R	Ratio
$E_{D/R,t}$	Benchmark rate, destination currency per unit of R	Ratio
$P^{\text{mid}}_{a,t}$, $P^{\text{ask}}_{a,t}$	Mid and ask price of S at the origin exchange	Origin currency
$P^{\text{mid}}_{b,t}$, $P^{\text{bid}}_{b,t}$	Mid and bid price of S at the destination exchange	Destination currency
$\rho_{a,t}$, $\rho_{b,t}$	Premium at the origin, destination exchange	Proportional
$\delta_{a,t}$, $\delta_{b,t}$	Half-spread at the origin, destination exchange	Proportional
$E_{D/O,t}$	Implied benchmark rate, $E_{D/R,t}/E_{O/R,t}$	Ratio
C	Cost of a transfer	Proportional
Panel D. Fees		
$f^{\text{dep}}_{a,m,\bar{t}}$, $F^{\text{dep}}_{a,m,\bar{t}}$	Fiat deposit fee at origin exchange, method m	Origin currency
$\tau_{a,\bar{t}}$	Trading fee at the origin exchange	Proportional
$g^{\text{wd}}_{a,n,\tilde{t}}$, $G^{\text{wd}}_{a,n,\tilde{t}}$	Stablecoin withdrawal fee at origin exchange, network n	Stablecoin
$g^{\text{dep}}_{b,n,\tilde{t}}$, $G^{\text{dep}}_{b,n,\tilde{t}}$	Stablecoin deposit fee at destination exchange, network n	Stablecoin
$\tau_{b,\bar{t}}$	Trading fee at the destination exchange	Proportional
$f^{\text{wd}}_{b,m,\bar{t}}$, $F^{\text{wd}}_{b,m,\bar{t}}$	Fiat withdrawal fee at destination exchange, method m	Destination currency
Panel E. Time indices		
t	Continuously determined market prices	—
$\bar{t}$	Administered fees, fixed in the short run	—
$\tilde{t}$	Fees that may vary with network conditions or be fixed	—

Notes: Balances are indexed by the stage of the transfer, following Table 1.

Table A2: An Illustrative Transfer of 200 Units

Panel A. Assumed rates, fees, and wedges

Stage	Parameter	Value	Denomination
—	Benchmark rate, origin, $E_{O/R,t}$	1.0000	O per unit of R
—	Benchmark rate, destination, $E_{D/R,t}$	4.0000	D per unit of R
1	Fiat deposit fee, $f^{\text{dep}}_{a,m,\bar{t}}$	0.25%	Proportional
1	Fiat deposit fee, $F^{\text{dep}}_{a,m,\bar{t}}$	0.0000	Origin currency
2	Premium, $\rho_{a,t}$	0.80%	Proportional
2	Spread, $\delta_{a,t}$	0.10%	Proportional
2	Trading fee, $\tau_{a,\bar{t}}$	0.40%	Proportional
3	Stablecoin withdrawal fee, $g^{\text{wd}}_{a,n,\tilde{t}}$	0.00%	Proportional
3	Stablecoin withdrawal fee, $G^{\text{wd}}_{a,n,\tilde{t}}$	0.5000	Stablecoin
3	Stablecoin deposit fee, $g^{\text{dep}}_{b,n,\tilde{t}}$	0.00%	Proportional
3	Stablecoin deposit fee, $G^{\text{dep}}_{b,n,\tilde{t}}$	0.0000	Stablecoin
4	Premium, $\rho_{b,t}$	0.30%	Proportional
4	Spread, $\delta_{b,t}$	0.10%	Proportional
4	Trading fee, $\tau_{b,\bar{t}}$	0.40%	Proportional
5	Fiat withdrawal fee, $f^{\text{wd}}_{b,m,\bar{t}}$	0.25%	Proportional
5	Fiat withdrawal fee, $F^{\text{wd}}_{b,m,\bar{t}}$	2.0000	Destination currency

Panel B. The balance through the chain

Stage	Balance and calculation	Value	Denomination	In R	Loss
0	$B_{0,O}$	200.00	O	200.00	—
1	$X_{1,O} = 200 \times (1 - 0.0025)$	199.50	O	199.50	0.50
2	$Q_{2,S} = 199.5/(1 \times 1.008 \times 1.001 \times 1.004)$	196.93	S	196.93	2.57
3	$Q_{3,S} = 196.93 - 0.5$	196.43	S	196.43	0.50
4	$X_{4,D} = 196.43 \times 4 \times 1.003 \times 0.999 \times 0.996$	784.14	D	196.04	0.39
5	$B_{5,D} = 784.14 \times (1 - 0.0025) - 2$	780.18	D	195.05	0.99
	Total loss				4.95
	Cost of the transfer, $C = 4.95/200$				2.47%

Notes: The table sets out the stylized transfer presented in Section 3.8. Panel A lists the assumed benchmark rates, fees, and wedges, including those set to zero; symbols carry the same subscripts as in Table A1. In Panel B, balances are shown in the currency in which they are held and converted into reference-currency units at the benchmark rates, so that losses across stages are comparable.

Table A3: Alternative Fiat Methods Used

	Mean	SD	Min	Median	Max	N corr.	N routes
Panel A. Total corridor cost (%)							
Cheapest (baseline)	2.18	1.84	−2.18	1.99	8.03	270	4,005
Cheapest, imputed	2.18	1.84	−2.18	1.99	8.03	270	4,028
Median	3.04	2.47	−2.18	2.46	10.97	270	4,005
Most expensive	4.07	3.54	−2.18	3.14	17.81	270	4,005
Panel B. Fiat rail cost (%)							
Cheapest (baseline)	0.67	1.04	0.00	0.31	4.86	270	4,005
Cheapest, imputed	0.67	1.04	0.00	0.31	4.86	270	4,028
Median	1.54	1.85	0.00	0.66	8.92	270	4,005
Most expensive	2.57	3.14	0.00	1.18	15.78	270	4,005

Notes: Data based on a $200 transfer. Each row reports corridor-level summary statistics under the indicated fiat fee selection rule. Panel A shows the total corridor cost, while Panel B shows the fiat part of the total cost. The baseline selects the cheapest available fiat deposit and withdrawal method per exchange. "Cheapest, imputed" additionally includes corridors where fiat fees are missing ("gate") but can be imputed from peer exchanges in the same country. "Median" and "Most expensive" replace the cheapest method with the median and most expensive observed method, respectively, for each exchange. The sample is locked to the 4,005 baseline-complete routes for all variants except "Cheapest, imputed." All values are expressed as a percentage of the transfer amount.
Source: Authors' calculations.

Table A4: Alternative Weighting Schemes for Aggregating Routes to Corridors

	Mean	SD	Min	Median	Max	N
Trade-count (baseline)	2.18	1.84	−2.18	1.99	8.03	270
Volume-weighted	2.33	1.92	−2.17	2.12	8.77	270
Equal exchange weights	2.33	1.72	−1.92	2.19	7.38	270
Equal exchange/coin/chain	2.22	1.71	−1.88	2.09	7.35	270

Notes: Data based on a $200 transfer. Each row reports corridor-level summary statistics under the indicated route weighting scheme. The baseline uses exchange trade-count weights within each corridor, combined with manual coin and chain weights. "Volume-weighted" replaces trade-count exchange weights with volume-based weights. "Equal exchange weights" assigns uniform weight to each exchange within a corridor, retaining the manual coin and chain weights. "Equal exchange/coin/chain" assigns uniform weight across all exchange, coin, and chain dimensions within each corridor. All values are expressed as a percentage of the transfer amount.
Source: Authors' calculations.

Table A5: Different Transfer Amounts

	Mean	SD	Min	Median	Max	N
$50	4.12	2.57	−2.04	3.72	11.40	270
$200 (baseline)	2.18	1.84	−2.18	1.99	8.03	270
$500	1.79	1.79	−2.21	1.58	7.81	270
$1,000	1.66	1.78	−2.22	1.45	7.75	270
$5,000	1.54	1.77	−2.23	1.25	7.70	270

Notes: Each row reports corridor-level summary statistics for the indicated transfer amount. For each variant, the cheapest fiat deposit and withdrawal method is re-selected at the relevant transfer amount, since the ranking of methods by cost may differ across transfer amounts when methods combine percentage and nominal fee components. All other inputs are held constant. All values are expressed as a percentage of the transfer amount for that variant.
Source: Authors' calculations.

Table A6: Premium Sensitivity

	Mean	SD	Min	Median	Max	N
Panel A. Baseline						
Median premium	2.18	1.84	−2.18	1.99	8.03	270
Panel B. Alternative premium scenarios						
No premium	2.21	1.05	0.40	2.06	6.18	270
25th percentile	2.20	1.39	−1.41	2.12	6.28	270
75th percentile	2.15	2.46	−3.35	1.93	9.25	270

Notes: Panel A uses the median premium for each exchange-stablecoin pair, matching the main results. Panel B varies the premium: the no-premium row sets the premium to zero on both legs, equivalent to using the benchmark exchange rate directly; the 25th (75th) percentile row uses the 25th (75th) percentile of the premium distribution.
Source: Authors' calculations.

Table A7: Corridor Cost by Country-Pair Group

	Mean	SD	Min	Median	Max	N
Panel A. Corridors (baseline)						
AE → AE	2.31	1.07	−0.12	2.37	4.54	42
AE → EMDE	1.61	1.07	−1.24	1.63	4.53	77
EMDE → AE	3.05	2.08	−0.85	2.52	8.03	61
EMDE → EMDE	2.01	2.22	−2.18	1.54	8.01	90
Panel B. Corridors (zero premium)						
AE → AE	2.32	0.38	1.39	2.28	2.90	42
AE → EMDE	2.00	0.74	1.02	1.80	4.08	77
EMDE → AE	2.72	1.17	1.36	2.37	6.18	61
EMDE → EMDE	2.01	1.26	0.40	1.55	5.34	90

Notes: Countries are classified as advanced economies (AEs) or emerging market and developing economies (EMDEs) following the IMF World Economic Outlook. Panel A reports the baseline scenario; corridors are separated on country clarification. Panel B assumes premium to be zero.

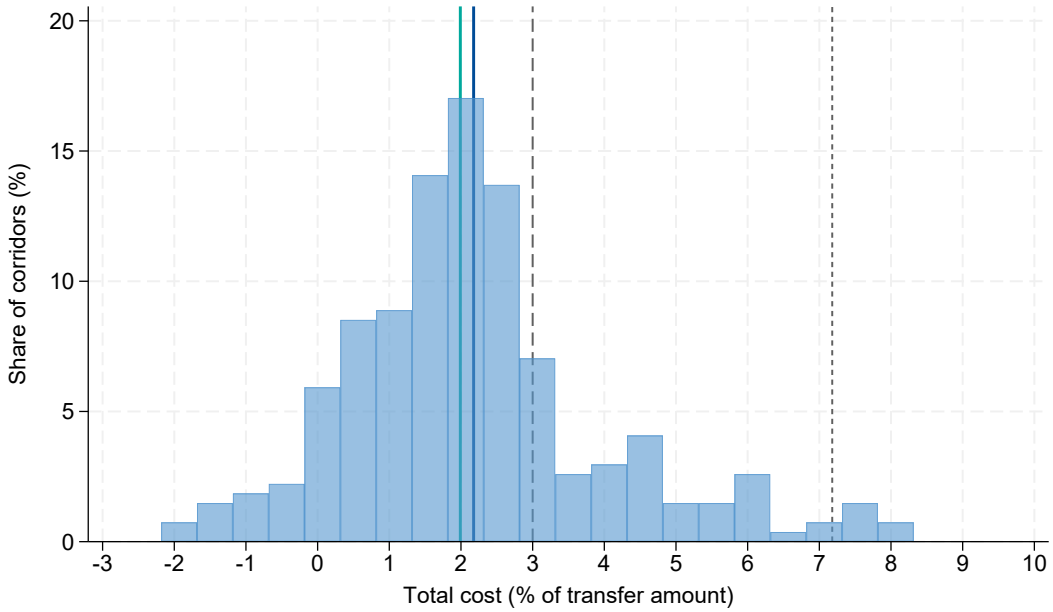


Figure A1: Corridor Cost Distribution

Notes: Data based on a $200 transfer. Each bar represents the share of corridors within a 0.5 percentage point cost bin. The vertical lines indicate the sample mean (dark blue), sample median (teal), and the G20 target of 3 percent and the global average of traditional remittance corridor costs (see Section 5.2 for more details) as dashed lines.
Source: Authors' calculations.

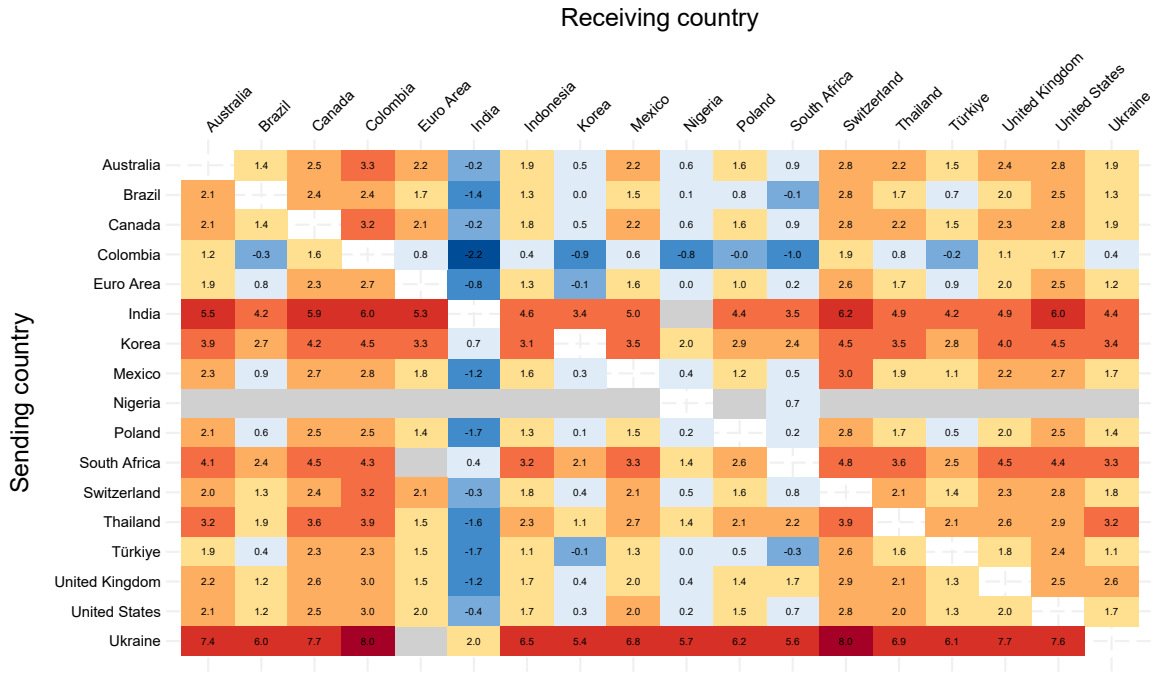


Figure A2: Corridor Cost Matrix

Notes: Each cell shows the cost of a $200 transfer for the corridor from the origin country (row) to the destination country (column), expressed as a percentage of the transfer amount. The color scale is diverging: blue shades indicate low or negative costs, transitioning through yellow to red shades for high costs. Diagonal empty cells are domestic transfers and are excluded from the analysis. Grey cells correspond to directed country pairs with no complete routes.
Source: Authors' calculations.

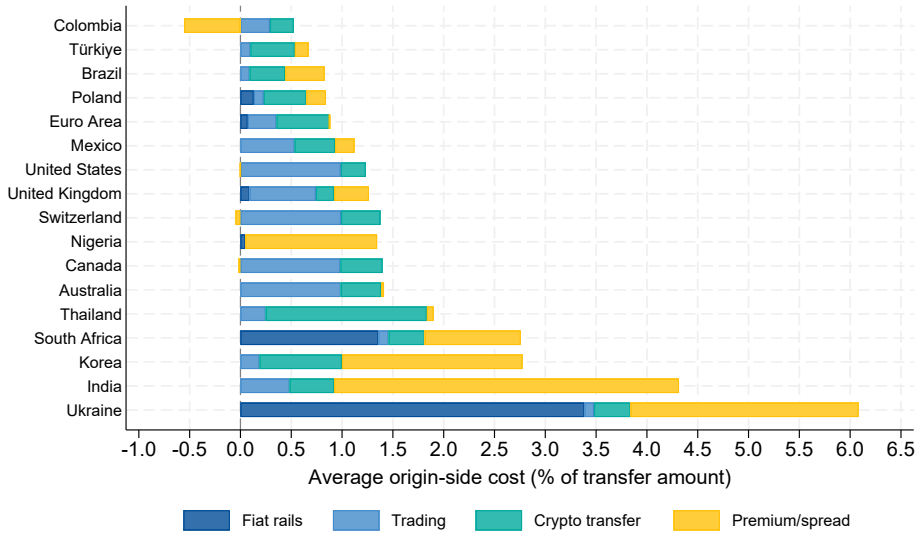


Figure A3: Sending Country Cost Decomposition

Notes: Data based on a $200 transfer. Each bar shows the average origin-side transfer cost for corridors where the indicated country is the sender, decomposed into four components: fiat deposit fees, exchange trading fees, crypto withdrawal fees, and market pricing frictions (premium and spread). Country averages are computed as the simple mean of weighted corridor costs across all corridors where that country appears as origin. Countries are sorted ascending by total origin-side cost. All values expressed as a percentage of the transfer amount.
Source: Authors' calculations.

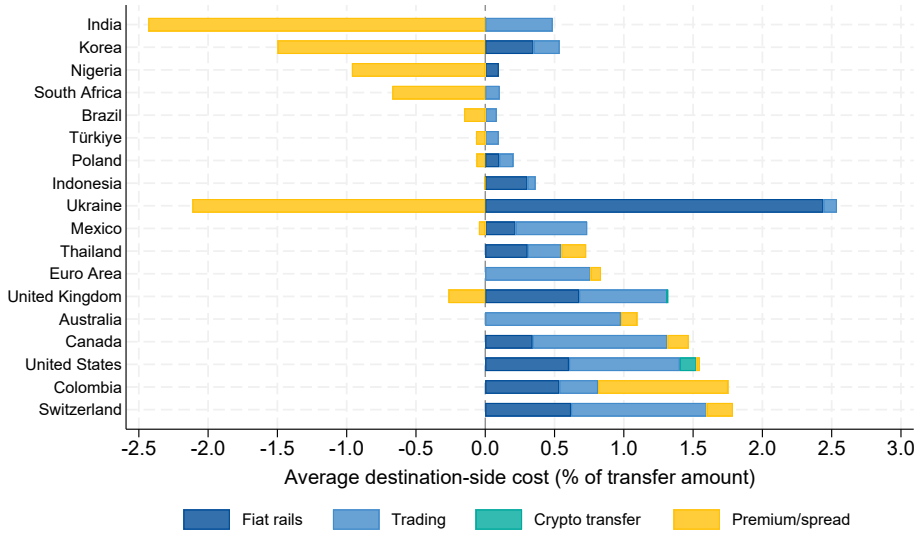


Figure A4: Receiving Country Cost Decomposition

Notes: Data based on a $200 transfer. Each bar shows the average destination-side transfer cost for corridors where the indicated country is the receiver, decomposed into four components: crypto deposit fees, exchange trading fees, fiat withdrawal fees, and market pricing frictions (premium and spread). A negative total reflects corridors where the destination-side premium is large enough to more than offset all destination-side fees. Country averages are computed as the simple mean of weighted corridor costs across all corridors where that country appears as destination. Countries are sorted ascending by total destination-side cost. All values expressed as a percentage of the transfer amount.
Source: Authors' calculations.

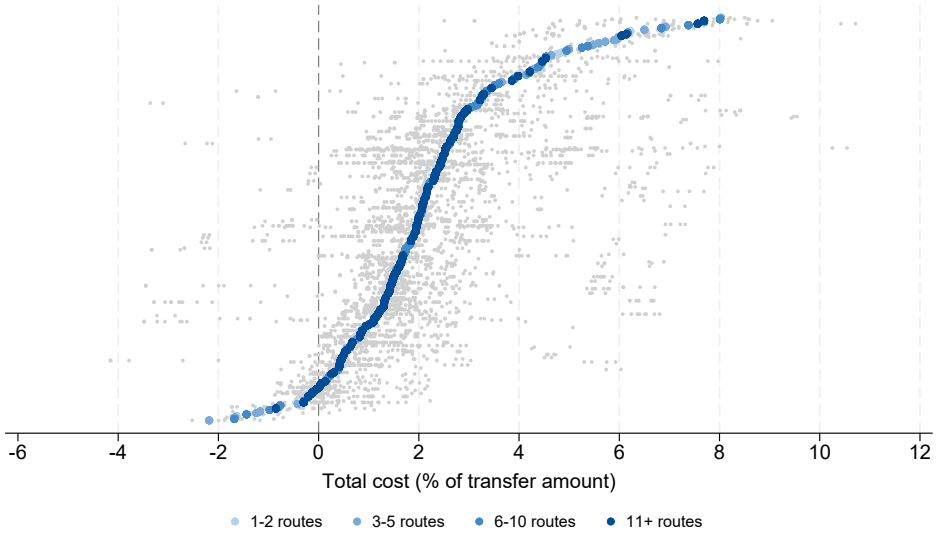


Figure A5: Routes Dispersion Around Corridor Mean

Notes: Data based on a $200 transfer. Each grey dot represents an individual complete route, plotted against the corridor's position (sorted ascending by weighted-average corridor cost). Routes within the same corridor are jittered vertically. The colored dot shows the weighted-average corridor cost; color indicates the number of complete routes in that corridor: lightest blue = 1–2 routes, darkest blue = 11 or more routes. All values expressed as a percentage of the transfer amount. The sample covers 270 corridors and 4,005 complete routes.
Source: Authors' calculations.

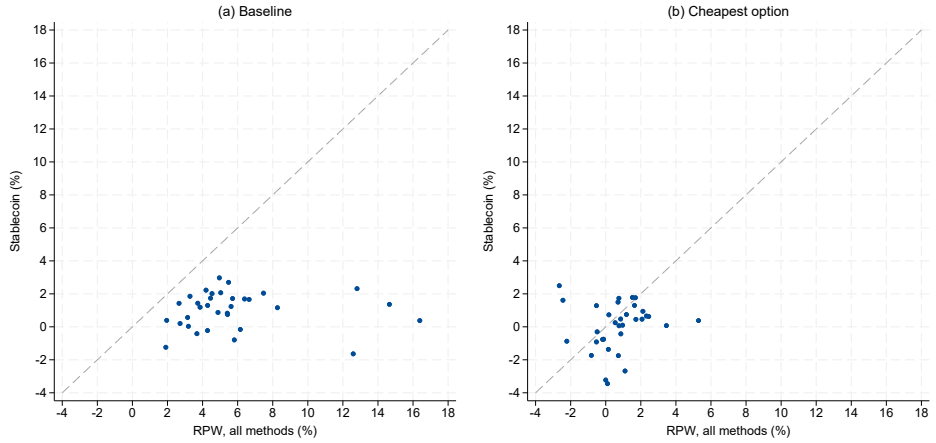


Figure A6: Comparing Stablecoin Corridor Costs to RPW Corridor Costs

Notes: Data based on a $200 transfer. Each dot represents one matched corridor. The left panel plots the stablecoin weighted-average corridor cost against the average cost across all traditional remittance providers in that corridor (RPW, all methods). The right panel plots the stablecoin cheapest route cost against the cheapest traditional remittance provider cost in that corridor. The 45-degree line indicates equal cost; dots below the line indicate corridors where stablecoin transfers are cheaper. The matched sample covers 33 corridors; Euro Area member states are treated as a single origin or destination. All values expressed as a percentage of the transfer amount.
Source: Authors' calculations.

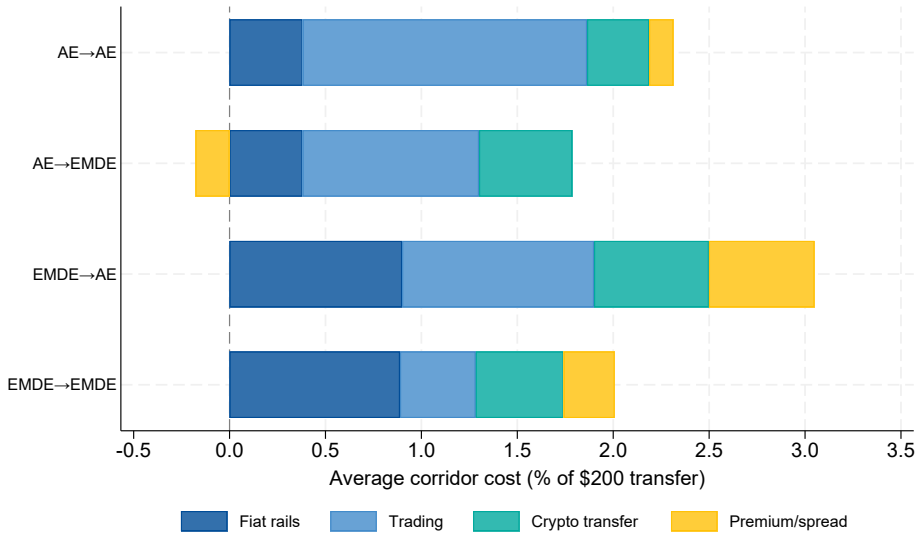


Figure A7: Decomposition of Average Corridor Cost by Country-Pair Group

Notes: Average total corridor cost, as a percent of a $200 transfer, decomposed into fiat rail fees, trading fees, crypto transfer costs, and the market premium and spread. Bars are equal-weighted means across corridors within each directed group. Classification follows the IMF World Economic Outlook.
Source: Authors' calculations.

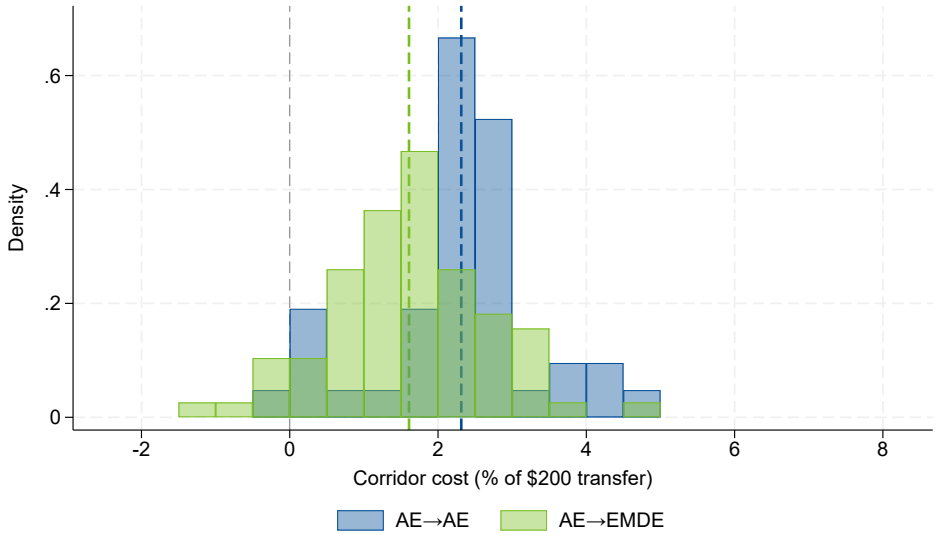


Figure A8: Distribution of Corridor Cost, AE-Sending Corridors

Notes: Histogram of total corridor cost, as a percent of a $200 transfer, for corridors originating in an advanced economy, split by whether the destination is an advanced economy (AE) or an emerging market and developing economy (EMDE). Vertical lines mark group means, and zero cost. Classification follows the IMF World Economic Outlook.
Source: Authors' calculations.

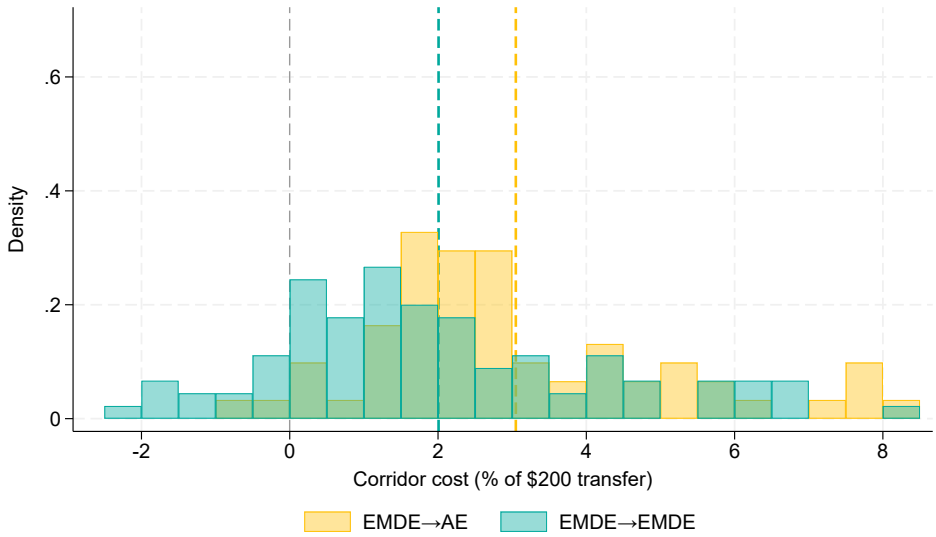


Figure A9: Distribution of Corridor Cost, EMDE-Sending Corridors

Notes: Histogram of total corridor cost, as a percent of a $200 transfer, for corridors originating in an emerging market and developing economy, split by whether the destination is an advanced economy (AE) or an emerging market and developing economy (EMDE). Vertical lines mark group means, and zero cost. Classification follows the IMF World Economic Outlook.
Source: Authors' calculations.

References

Aldasoro, I., Beltrán, P., and Grinberg, F. (2026). Stablecoin inflows and spillovers to FX markets. BIS Working Papers 1340, Bank for International Settlements. Also published as IMF Working Paper WP/26/056.

Auer, R., Lewrick, U., and Paulick, J. (2025). DeFiying gravity? An empirical analysis of cross-border Bitcoin, Ether and stablecoin flows. BIS Working Papers 1265, Bank for International Settlements.

Cardozo, P., Fernández, A., Jiang, J., and Rojas, F. D. (2024). On cross-border crypto flows: Measurement, drivers and policy implications. IMF Working Paper 2024/261, International Monetary Fund.

Choi, K. J., Lehar, A., and Stauffer, R. (2022). Bitcoin microstructure and the kimchi premium. Available at SSRN: https://ssrn.com/abstract=3189051.

Corwin, S. A. and Schultz, P. (2012). A simple way to estimate bid-ask spreads from daily high and low prices. *The Journal of Finance*, 67(2):719–760.

CPMI (2018). Cross-border retail payments. Technical Report CPMI Papers No. 173, Bank for International Settlements, Basel, Switzerland.

Di Iorio, A., Di Stefano, E., Mascioli, M., and Trebeschi, G. (2026). Are stablecoins efficient for remittances? evidence from a mystery shopping exercise by banca d'italia. Markets, Infrastructures, Payment Systems 86, Banca d'Italia.

Du, W., Huang, C., and Scharfstein, D. (2026). Competing rails for cross-border payments: Banks, fintechs and stablecoins. Working paper, Harvard Business School.

Financial Stability Board (2025). G20 roadmap for cross-border payments: Consolidated progress report for 2025. Technical report, Financial Stability Board.

Graf von Luckner, C., Reinhart, C. M., and Rogoff, K. S. (2023). Decrypting new age international capital flows. *Journal of Monetary Economics*, 138:104–122.

Graf von Luckner, C. M., Koepke, R., and Sgherri, S. (2024). Crypto as a marketplace for capital flight. IMF Working Paper 2024/133, International Monetary Fund.

International Monetary Fund (2025). Understanding stablecoins. Departmental paper, International Monetary Fund.

International Monetary Fund (2026). Foreign currency stablecoins: Initial considerations on policies to manage and enhance macro-financial stability. Departmental paper, International Monetary Fund. Forthcoming.

Kang, C.-M., Kang, H.-G., Kim, D., and Sul, H. K. (2025). Stablecoin and cross-border crypto market integration. *Economics Letters*, 257.

Ma, Y., Zeng, Y., and Zhang, A. L. (2025). Stablecoin runs and the centralization of arbitrage. NBER Working Paper 33882, National Bureau of Economic Research.

Reuter, M. (2025). Decrypting crypto: How to estimate international stablecoin flows. IMF Working Paper 2025/141, International Monetary Fund.

Reuter, M. (2026). International stablecoin flows, capital controls and currency crises. IMF Working Paper 2026/XXX, International Monetary Fund. Forthcoming.

Watsky, C., Allen, J., Daud, H., Demuth, J., Little, D., Rodden, M., and Seira, A. (2024). Primary and secondary markets for stablecoins. *FEDS Notes*.

World Bank (2024). World development indicators. `https://databank.worldbank.org/source/world-development-indicators`. Accessed May 2026.

World Bank (2025). Remittance prices worldwide quarterly. Technical Report 54, World Bank. (Q3 2025).



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